Apparent minimality violations solved by Nested Agree

Irene Amato

Humboldt-Universität zu Berlin

December 6, 2024

Minimality is a well-established locality condition on syntactic dependencies. Nonetheless, minimality violations are attested in different domains and different languages. One phenomenon that seems to violate minimality is auxiliary selection in Standard Italian. In this language, the morphological realization of the perfect auxiliary depends on the features of the lower arguments, rather than on the higher subject. I propose that such configurations can be accounted for by *Nested Agree*. This principle prescribes that an already established dependency between two heads must be exploited by all the features located on those heads. The novelty of Nested Agree compared to similar principles lies in the assumption of feature ordering, which makes this principle not transderivational, not specificity-based, and able to model cases of anti-minimality and anti-locality. In particular, an alleged intervener lies outside the search domain of a probe feature F on a head H, if the search domain of F has been restricted by a previous operation carried out by the head H. I will show that this is exactly what happens in Italian auxiliary selection. In addition, I discuss several phenomena as further support for Nested Agree: case and agreement alignments in ditransitive clauses, dative intervention in Icelandic, subject agreement in VOS clauses in Spanish, person agreement in Lak, multiple wh-fronting in Bulgarian.

Keywords: minimality, locality, Agree, feature ordering, auxiliary selection, Standard Italian

1 Introduction

Minimality is a computational requirement that reduces the search space of an operation up to the highest matching goal. Formulations of this principle are, for example, *Relativized Minimality* (Rizzi 1990) and *Minimal Link Condition (MLC)* (Fanselow 1991; Chomsky 1995b, 2001). Although it is well-established that syntactic operations are subject to locality conditions of this type, anti-minimality effects are attested in different domains and different languages. The main case study presented in this paper is auxiliary selection in Standard Italian. This phenomenon challenges minimality because the morpho-phonological realization of the perfect auxiliary depends on the features of the lower arguments, rather than on the higher subject. For instance, transitive clauses require the auxiliary HAVE, but if they contain an indirect reflexive object the auxiliary is instead BE.

In this paper, I introduce a new way to deal with anti-minimality configurations and intervention effects in syntactic dependencies. Following and further developing Amato (2023), I introduce the principle *Nested Agree*. Assuming that the features located on the same head are extrinsically ordered, Nested Agree prescribes that an already established dependency between two heads must be exploited by all the features located on those heads. As a consequence, a probe initiating an operation after another probe located on the same head should pick out the same goal as the preceding probe did. Nested Agree affects the search domain of ordered operations: if an operation is triggered by a head that has already carried out another operation, the previous one restricts the search domain of the subsequent one. Under this perspective, minimality violations do not arise (despite superficial appearance) if the higher, alleged intervener actually lies outside the search domain of the probe, as determined by previous operations. Hence, the sequence of derivational steps may produce a representation in which minimality is opaque, even though it is indeed respected during the application of each operation.

Nested Agree allows a precise and comprehensive analysis of auxiliary selection in Standard Italian, which seems to be impossible without this principle. I consider auxiliary selection as the result of Agree for the person feature. The minimality violation given by the dependence of the auxiliary on the features of the lower argument is only apparent: the higher subject actually lies outside the search domain of the relevant probe, and the lower argument is the closest possible goal.

Besides auxiliary selection, I also provide other case studies where Nested Agree seems to be the right principle to handle the data: case and agreement alignments in ditransitive clauses, dative intervention in Icelandic, subject agreement in VOS clauses in Romance languages, person agreement in Lak, multiple wh-fronting in Bulgarian. In addition, I also discuss how the action of Nested Agree can be restricted, showing that it can exclude impossible patterns, independently of the assumed feature ordering.

The paper is structured as follows. In section 2, I introduce the principle *Nested Agree*. I analyze auxiliary selection in section 3, which contains the data (3.1), an overview of previous analyses (3.2), the proposal (3.3), the relevant derivations (3.4), and a short discussion of cross-linguistic variation (3.5). In section 4, I offer several case studies as further support for Nested Agree: the impossibility of secundative case and indirective agreement in ditransitive clauses (4.1.1), dative intervention in biclausal sentences in Icelandic (4.1.2), ϕ-agreement in VOS clauses in Spanish (4.2.2), person agreement in Lak (4.2.1), and multiple wh-fronting in Bulgarian (4.2.3). I also show that potential counterexamples can be explained in various ways: anti-local agreement in Hindi-Urdu (4.3.1), and participle agreement in Italian, Passamaquoddy, and Dinka (4.3.2). I conclude in section 5. The status of Nested Agree with respect to other principles already proposed in the literature is discussed in appendix 6.

2 Nested Agree

In this section, I introduce the principle *Nested Agree*. The case studies addressed in sections 3 and 4 will provide support for this principle. The adopted frameworks are *Minimalism* (Chomsky 1995b, 2000, 2001) and *Distributed Morphology* (Halle & Marantz 1993; Harley & Noyer 1999). Grammar is considered a modular, derivational model, where all operations obey strict cyclicity.

Under the assumption that operations triggered by the same head are ordered (Koizumi 1994; Sabel 1998; Heck & Müller 2006; Müller 2009; Georgi 2014; Assmann et al. 2015), I propose that the operation that comes next must target the same goal that the previous operation has targeted. I call this principle *Nested Agree*, defined in (1).[1]

[1]This insight is already present in Amato (2022, 2023), which are predecessors for this principle. In this article,

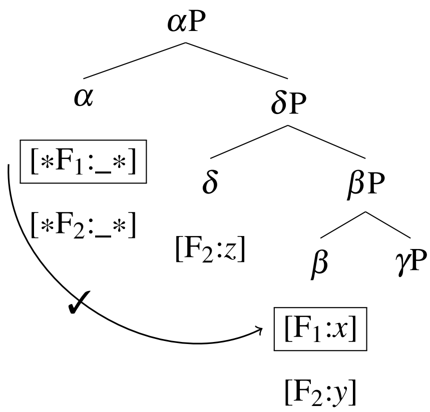
(1) *Nested Agree*

Let $*F_1*$ and $*F_2*$ be two ordered probes on the same head H. If the operation A_1 for the feature $*F_1*$ has targeted the head G, then the subsequent operation A_2 for the feature $*F_2*$ must also target G.

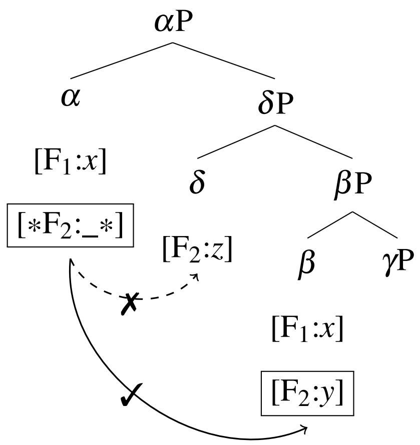
Nested Agree prescribes that a probe-feature ($*F_2*$) initiating an operation after another probe-feature ($*F_1*$) located on the same probe-head (H) should pick out the same goal-head (G) as the preceding probe-feature ($*F_1*$) did. The central idea of Nested Agree is that an "Agree-channel" that has been opened by a previous dependency must be reused for a different probing operation, minimizing the number of independent Agree relations. The denomination *nested* indicates that the domain of a subsequent operation is a smaller subset of the domain of the previous operation, as in a matryoshka doll.

The Nested Agree configuration is represented in trees (2) and (3).

(2) *Agree for [$*F_1$:_$*$]*



(3) *Agree for [$*F_2$:_$*$]*



The feature ordering on α is $[*F_1*] \succ [*F_2*]$. The goal-head β is targeted twice by the probe-features on α: firstly by $[*F_1*]$ in (2), and secondly by $[*F_2*]$ in (3). Although δ is a closer matching goal-head for the probe $[*F_2*]$ on α, it lies outside the search domain of $[*F_2*]$ because every feature on α must target the same goal-head, exploiting the dependency opened by $[*F_1*]$. Hence, Nested Agree allows us to treat minimality violations from a new perspective. An apparent intervener δ can be circumvented if the non-local goal β is an item that has already

I offer a more minimal version of *Nested Agree*, apply it to further scenarios, including the operation Merge as well, and argue that this principle constitutes a new tool to deal with minimality violations in general.

been involved in an Agree relation with the head α. Despite its lower position, [F_2] on β is actually the closest possible goal for [$*F_2*$] on α. The relations represented in trees (2) and (3) are all local, but the sequence of operations has produced a representation in which locality is opaque.

If the goal β does not bear the feature [F_2], several scenarios could occur: either Agree for [$*F_2*$] fails, or a repair strategy applies. Possible repairs are *cyclic expansion* à la Béjar & Rezac (2009), or *downward expansion* without backtracking. In the former, the unsatisfied probe can initiate a second cycle of Agree after the search space is enlarged by Merge. In the latter, the probe starts its downward search exactly from the goal favored by Nested Agree. The search stops when the whole syntactic structure has been inspected, or when a phase boundary is encountered (as prescribed by the *Phase Impenetrability Condition (PIC)*; Chomsky 2000: 108, Chomsky 2001: 13).

Nested Agree can be formulated with reference to *Agree-Link* (Arregi & Nevins 2012). Under this theory, Agree is split into two operations: *Agree-Link* and *Agree-Copy*. The former creates the syntactic dependency between two heads (see also Bjorkman & Zeijlstra 2019; Baker & Souza 2020). Syntax stores the link between the probe-head and the goal-head, and not only the relation between the probe-feature and the goal-feature. Within this theory, the principle of Nested Agree can be stated as follows: given two ordered probes [$*F_1*$] $\succ$ [$*F_2*$], the Agree-Link established for the feature [$*F_2*$] must be, if possible, the Agree-Link already established for the feature [$*F_1*$].[2]

The main novelty of Nested Agree with respect to similar principles stems from the background assumption that features located on the same head are extrinsically ordered (see appendix 6 for discussion). The idea that syntactic elements contain sequences of features to be checked in a certain order can already be found in Chomsky (1995b: 179), as highlighted by Sabel (1998: 134-135). In general, operations triggered by features that enter the derivation at the same time (i.e., on the same head) can apply either simultaneously or sequentially. Both the simultaneous

[2]The Agree-Link theory can be used to justify the principle of Nested Agree because it offers an independent motivation to assume that there is a formal link that exists independently from the actual valuation procedure. Once the two heads H and G enter a relation for a feature $*F_1*$, an Agree-Link between H and G is created. Note that the existence of such a link does not guarantee that another probe-feature on the same probe-head must, or even can, exploit this previous Agree-Link. This is the new part introduced by Nested Agree.

and sequential approaches require assumptions regarding the structure of syntactic elements and operations. I adopt the sequential view because it fits a strictly derivational view of syntax. If only a single operation can apply at any given stage of the derivation, then features on the same head have to be ordered. Evidence for postulating a specific feature ordering comes from two aspects: the interactions between different operations (cases of feeding and bleeding), and the existence of a typology of languages built on different feature orderings (see sections 3.5, 4.1.1 for modeling of different grammars via feature re-ordering). If the analyses based on feature ordering have adequate empirical coverage both within a language and cross-linguistically, as seems to be the case, then feature ordering is a valid and useful assumption.[3]

3 Nested Agree and auxiliary selection in Standard Italian

3.1 The distribution of HAVE and BE

The term *auxiliary selection* refers to the alternation between the auxiliaries BE and HAVE in the Romance and Germanic periphrastic perfect constructions (Perlmutter 1978; Burzio 1986; Kayne 1993; Sorace 2000; Bjorkman 2011, 2018). In some languages (such as German), the alternation depends on the argument structure (*argument-structure-based systems*); in other languages (such as many Southern Italo-Romance dialects), it is related to the person feature of the subject (*subject-driven systems*). In Standard Italian, auxiliary selection depends on the argument structure. With transitive verbs, the auxiliary is realized as HAVE (4-a).[4] As for

[3]Many analyses make explicit use of feature ordering, which is in general justified by their coverage of the data and/or by their typological predictions. Let me provide some examples. Assmann et al. (2015) derive the difference between nominative-accusative languages and ergative-absolutive languages by re-ordering Merge and Agree. For other analyses where Merge and Agree are ordered on a single head, see Bruening (2005); van Koppen (2005); Anand & Nevins (2006); Lahne (2008); Müller (2009); Asarina (2011); Halpert (2012); Richards (2013); Kalin & van Urk (2015); Kučerová (2016); Bjorkman (2018). Different dialects of Icelandic are modeled with different orders of operations (Agree on T and raising of dative experiencer) by Sigurðsson & Holmberg (2008). Feature ordering (of agreement features on C) is also used by Ershova (2024) to model islandhood in West Circassian; dialects that differ on this point have different feature orderings. Similarly, optionality within the same language is modeled with variable feature orderings by Puškar (2018). More in general, Manetta (2011) argues that bundles of features form ordered stacks. Martinović (2021) also transfers some aspects of the hierarchy of functional projections into a feature hierarchy on a head. In addition, feature ordering is also implicitly assumed for the mechanism of *phase unlocking* (see Rackowski & Richards 2005; Branan 2018, among many others). Some feature ordering might also be universal: according to Béjar & Rezac (2003); Preminger (2011), person always probes before number.

[4]Where not indicated differently, examples are mine. I am a native speaker of Standard Italian from the region of Lucca, Tuscany. Similar examples are well known from all the previous literature on Italian auxiliary selection (see,

intransitive verbs, the perfect auxiliary is HAVE with unergative verbs (4-b), BE with unaccusative verbs (4-c).[5]

(4) a. Teresa ha lavato le camicie.
Teresa have.PRS.3SG wash.PRTC the shirts
'Teresa has washed the shirts.'

b. Teresa ha risposto a sua madre.
Teresa have.PRS.3SG answer.PRTC at her mother
'Teresa answered her mother.'

c. Teresa è venut-a alla festa.
Teresa be.PRS.3SG come.PRTC-SG.F to.the party
'Teresa has come to the party.'

The perfect auxiliary is realized as HAVE only when transitive or unergative verbs select canonical arguments, as in (4-a)-(4-b). If one of the arguments is a reflexive or an impersonal pronoun, the auxiliary is instead realized as BE. The reflexive pronoun can be either the direct object (direct reflexive clauses), (5-a), or the indirect object (indirect reflexive clauses), (5-b)-(5-c). The impersonal pronoun *si* can appear with all types of predicates. An example with a transitive verb is given in (5-d).

(5) a. Teresa si=è lavat-a.
Teresa REFL.ACC.3SG=be.PRS.3SG wash.PRTC-SG.F
'Teresa washed herself.'

b. Teresa si=è lavat-a il vestito.
Teresa REFL.DAT.3SG=be.PRS.3SG wash.PRTC-SG.F the dress
'Teresa washed her dress (for herself).'

c. Teresa si=è rispost-a.
Teresa REFL.DAT.3SG=be.PRS.3SG answer.PRTC-SG.F

---

for instance, Burzio 1986: 53-55). I have chosen to provide my own data to have a simple and consistent paradigm for expository purposes.

[5]Some intransitive verbs may combine with both auxiliaries (i).

(i) Maria ha corso / è cors-a velocemente.
Maria have.PRS.3SG run.PRTC / be.PRS.3SG run.PRTC-SG.F fast
'Maria has run fast'. (Sorace 2000: 876)

This fact is orthogonal to the analysis illustrated in this paper. In particular, I assume that the lexical semantics of each verb determines its syntactic argument structure, and, consequently, the type of *v* head with which it combines. As we will see, it is the head *v* that conditions the morpho-phonological realization of the auxiliary. For instance, the verb *correre* 'run' can be selected either by an unaccusative *v* or by an unergative/transitive *v* (as evidenced by the possibility of PPs in the former, and overt objects in the latter), which results in the alternation in (i).

'Teresa has answered herself.'

d. Ieri si=è mangiato il gelato.
yesterday IMPERS=be.PRS.3SG eat.PRTC the ice cream
'Yesterday one has eaten the ice cream.'

These data show that in Italian not only the argument structure counts (cf. (4-a), (4-b) vs. (4-c)), but also the features of the arguments play a crucial role (cf. (4-a) vs. (5-a), (5-b), (5-d)). The relevant factors are the type of direct and indirect object (reflexive vs. non-reflexive), and the presence of an impersonal argument. Hence, the syntactic features that characterize reflexive vs. non-reflexive, impersonal vs. non-impersonal arguments must be responsible for the different morpho-phonological realizations of the perfect auxiliary.

3.2 Previous analyses of auxiliary selection

There are many different existing approaches to auxiliary selection, but none can fully capture the data described in section 3.1 and provide an explicit, derivational analysis of auxiliary selection.

Kayne (1993); Cocchi (1995) develop an incorporation analysis. Both possessive and perfect structures contain a dedicated syntactic head BE and an "abstract" nominal or prepositional head D/P. HAVE is the result of incorporation of the D/P head into the BE head. The first problem is that these analyses make use of no longer adopted assumptions that can be hardly adapted to current theories (but see Bjorkman 2011: 165-171 for discussion). The second issue concerns the D/P head: it does not contribute to semantic interpretation, and it must be absent in certain structures, such as unaccusative sentences. Thirdly, the empirical coverage for subject-driven auxiliary selection is not adequate: only the distribution BE-BE-HAVE can be derived, but many different patterns are attested (see section 3.5).

D'Alessandro & Roberts (2010: 51) argue for a rule of auxiliary selection: in Standard Italian, HAVE is inserted when *v* is non-defective, otherwise BE is chosen. Clearly, if the constraining factor is the argument structure itself, HAVE should show up with every transitive predicate, including (5-a), (5-b), (5-d). D'Alessandro & Roberts (2010: 51, fn.7) are aware of the problem of reflexives, but do not provide any solutions to solve it.

Bjorkman (2011, 2018) proposes an agreement-based analysis. HAVE appears when the verb selects an external argument (detected via Agree for a [D] feature), otherwise BE shows up. If it is the sole presence of the external argument that determines the form of the auxiliary, BE should not be expected in transitive reflexive clauses such as (5-a) and especially (5-b). Reflexive and impersonal clauses are not mentioned in Bjorkman (2011, 2018).

3.3 Auxiliary selection is Agree for the person feature

Since the morpho-syntactic features of the arguments are the relevant factor, I consider auxiliary selection as an instance of Agree for the person feature, following Amato (2022, 2023). The dependency between the perfect head and another head is modeled via *Agree*, which is the operation that transmits information across the syntactic structure. I adopt a standard definition of Agree: it is a downward search operation that requires feature matching, c-command, and locality (Chomsky 2000, 2001).

I assume that a probe is any feature that can initiate such a search operation, independently of its valuation. The activity of a feature is signaled by the diacritic $*$, which is erased after Agree: $[*\mathrm{F}*] \rightarrow [\mathrm{F}]$ (notation from Heck & Müller 2006). A probe can be either unvalued, as ϕ-probes are ($[*\phi{:}\_*]$), or valued, as case assigners are ([$*$case:acc$*$]).[6] Agree can fail as far as valuation is concerned (*failed Agree*, Preminger 2014). In addition, a probe can become a goal, after it has triggered its operation.

For the perfect auxiliary, I adopt a syntactic representation in line with D'Alessandro & Roberts (2010); Bjorkman (2011, 2018). The head Perf is the first temporal functional projection outside the *v*P (Bjorkman 2018: 330). It introduces the perfect semantics, it selects a *v*P and it is selected by T.

I propose that the head Perf bears a person probe $[*\pi{:}\_*]$, which is responsible for the interaction of the perfect auxiliary with both the argument structure and the features of the arguments. Whenever Agree for $[\pi]$ on Perf is successful, HAVE is inserted. If Agree fails, the

[6]The standard definition of *probe* states that this is an uninterpretable, unvalued feature. Here, I suggest that probes may also be valued. The idea of probes as uninterpretable but not necessarily unvalued features was originally proposed by Chomsky (2000). The dissociation between interpretability and valuation as two independent conditions goes back to Pesetsky & Torrego (2007). Note, however, that I consider *uninterpretability* as a syntactic property, rather than a semantic one: the uninterpretability of a feature, indicated by the diacritic $*$, means that the feature can trigger a search operation and has not done it yet.

elsewhere form BE must be chosen. Indeed, there must be a feature that determines auxiliary selection (another alternative is a [D] feature, see Kayne 1993; Bjorkman 2011, 2018). I suggest that this should be a π-feature for two reasons. Firstly, in Italian the auxiliary depends on the features of the arguments (reflexive vs. non-reflexive, and impersonal vs. non-impersonal arguments). This fact can be easily modeled by assuming that different arguments bear different sets of ϕ-features: defective vs. non-defective. Secondly, in those dialects where auxiliary selection is subject-driven, the auxiliary depends on the person feature of the subject (Loporcaro 2007; Ledgeway 2019).

Auxiliary selection is the result of Agree between Perf and the person feature located on the lower head *v* (valued in a prior Agree operation between *v* and the internal argument). Auxiliary selection is due to *cyclic Agree* (Legate 2005) between Perf and a lower argument, enabled by the intermediate head *v*, as shown in (6).

(6) $\underbrace{\text{Perf}[*\pi{:}\_*] \text{ ...... } \text{DP}_{\text{sbj}}[\pi{:}\beta] \text{ ...... } v}_{\pi\text{-Agree}}\underbrace{[*\pi{:}\_*] \text{ ...... } \text{DP}_{\text{obj}}}_{\pi\text{-Agree}}[\pi{:}\alpha]$ ......

However, a careful observation of the operation described in (6) reveals that this Agree configuration should not be possible, since it creates a non-local dependency. The person feature on *v* is not local to the π-probe on the perfect head.[7] At least the subject (and eventually other items, such as clitics or wh-phrases) should be closer to Perf. Thus, auxiliary selection in Standard Italian poses a problem of minimality.

Nested Agree provides a solution to this puzzle. Crucially, the two heads Perf and *v* are also involved in another operation besides Agree for person. In the perfect periphrasis, the lexical verb appears in the form of a past participle. I suggest that this is the effect of an inflectional feature [Infl] located on the head Perf (see Bjorkman 2011 for [Infl]). Perf bears a valued probe [∗Infl:perf∗], which needs to be checked with a feature of the same type, located on *v*. In fact, it

[7]This hinges on the assumption that there are no π-features on the *v*P-level. Only categorical, inherent features project onto maximal projections, whereas other features, such as ϕ-features acquired throughout the derivation, do not project. The idea that not every feature of the head projects onto the phrasal node is already envisaged in Chomsky (1995a: 398), according to which the label and the head are connected, "though not always through strict identity". Similar suggestions have been made by Vicente (2007); Harizanov & Gribanova (2019), among others. The question of whether (discharged) π-probes project is, at least partially, an empirical one. In the present study, the projection of the π-probe on the *v*P level leads to wrong predictions about unaccusative clauses, subject-driven auxiliary selection, and subject agreement (see footnote 9). Thus, Italian does not provide evidence for postulating feature projection of (discharged) probes.

is the higher aspectual/temporal head Perf that determines the form of the lexical verb. Hence, the item that induces the Agree operation is the higher one.[8] The valued probe [$*$Infl:perf$*$] searches downwards for a matching feature. It finds [Infl:_] on *v*, which acquires the value *perf*. Because of [Infl:perf], the verb will be morphologically realized as a past participle.

In Standard Italian, the features on Perf are ordered as in (7).

(7) [$*$Infl:perf$*$] $\succ$ [$*\pi$:_$*$] (argument-structure-driven systems)

According to (7), the Infl-probe acts before the π-probe. The first Agree operation for [Infl] creates a dependency between *v* and Perf, shown in (8).

(8) Perf[$*$Infl:perf$*$][$*\pi$:_$*$] ... DP$_{\text{sbj}}$[π:β] ... *v*[Infl:_][π:α] ... DP$_{\text{obj}}$[π:α] ...

Infl-Agree

The operation in (8) opens a "channel" that must be reused for person Agree, as established by Nested Agree. Hence, Perf agrees with *v* for person, thereby skipping the subject, as was shown in (6). There is no minimality violation in (6) because π-Agree on Perf makes use of the already established well-formed Agree dependency in (8) (between Perf and *v* for [Infl]), which bypasses the offending intervening feature (on the DP$_{\text{sbj}}$).

Nested Agree is necessary to derive the distribution of the perfect auxiliaries in Standard Italian. There are at least two contexts where this principle is clearly needed. Firstly, the person probe on Perf must get past the subject of transitive verbs and agree with *v*, as illustrated in (6). This dependency must take place because the auxiliary must be realized as BE in transitive clauses with reflexive internal arguments (cf. section 3.4.3). Secondly, the raised internal argument of unaccusative verbs must be skipped by the person probe on Perf to account for BE with unaccusative verbs (cf. section 3.4.2). Note that standard Agree mechanisms cannot handle these data because they do not allow any higher argument to be skipped, in accordance with minimality. Similarly, principles like *Maximizing Matching Effects* (Chomsky 2001), which imposes the satisfaction of all probes at once among matching heads, or the *Earliness principle* (Pesetsky 1989), which prescribes that an Agree operation between a probe and a goal should

[8] For a similar inflectional dependency, see the German *status government* (Bech 1955); for a similar mechanism to implement morphological selection in syntax, see Wurmbrand (2012).

happen as soon as possible, cannot account for such cases, where probing is not maximized and does not involve the higher DP. In general, without adding the assumption of feature ordering, which is the core of Nested Agree, none of the previously proposed mechanisms can handle such cases, since minimality always relegates probing to the higher element.[9] Alternative Agree mechanisms and the novelty of Nested Agree with respect to previous approaches will be further discussed in the appendix 6.

As a further advantage, Nested Agree allows us to model subject-driven auxiliary selection with different extrinsic feature orderings on the head Perf, providing a unified analysis for these two very different systems of auxiliary selection found in extremely closely related languages (see section 3.5).

3.4 Derivations

3.4.1 Transitive clauses

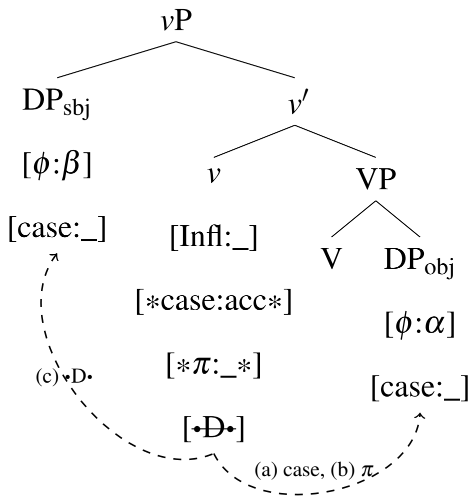
In Standard Italian, transitive clauses require the perfect auxiliary HAVE (cf. (4-a)). I now propose the derivation for this type of clause.[10] First of all, the VP is built and merged with a transitive *v*. I consider transitive *v* to be a full argument structure *v*, which assigns accusative

[9]No other principle can solve this minimality problem. *Tucking-in* (Mulders 1997; Richards 1997) might constitute a solution against anti-minimality for movement. However, its application to Agree is not straightforward. It could eventually give rise to Multiple Agree with both goals (including the non-local goal). If Perf copies the features of both the object and the subject, one would need either an additional system of indices to distinguish the different sources of the features, or an "overwriting" mechanism that favors the features of the lower argument. *Equidistance* (Chomsky 2001) does not help either, since it is defined for multiple specifiers and not for a head and its specifier (as the DP_{sbj} and *v* are). The *A-over-A condition* (Chomsky 1973; Hornstein 2009; Roberts 2010) may derive the dependency between Perf and *v* in (6) if taken in the following specific interpretation: if a probe can be checked either with a head, or with an edge element, it must be checked with the head (Müller 2004: 19). However, this version of A-over-A minimality leads to some wrong predictions. Firstly, it predicts that the auxiliary of unaccusative predicates is HAVE (cf. the derivation of unaccusative clauses in section 3.4.2). If the π-probe on Perf first targets the maximal projection *v*P, when the head *v* does not contain any person feature (i.e., when *v* is unaccusative) the probe goes on searching and targets the unaccusative subject in Spec,*v* (this DP must move to Spec,*v* because *v* is a phase; even if this assumption is rejected, the probe targets this DP in the internal argument position). This leads to successful agreement in unaccusative clauses, causing HAVE insertion. Secondly, this analysis can be hardly adapted to cross-linguistic variation: this principle should not be active in subject-driven systems, where the person probe on Perf targets the subject in Spec,*v* rather than *v*. Thirdly, this version of A-over-A minimality predicts that languages like Italian and English should have object agreement on T in transitive clauses (i.e., T should agree with *v*P rather than with the subject).

[10]I propose the very same analysis for unergative verbs. I assume that unergative verbs are universally underlyingly transitive (see Laka 1993; Hale & Keyser 1993, 2002; Pineda 2014; Baker & Bobaljik 2017, among others). Unergative verbs merge with an object with the same lexical features of the verb, but with no phonological realization. Crucially, I assume that the covert cognate object bears syntactic features valued as default (i.e., third person singular).

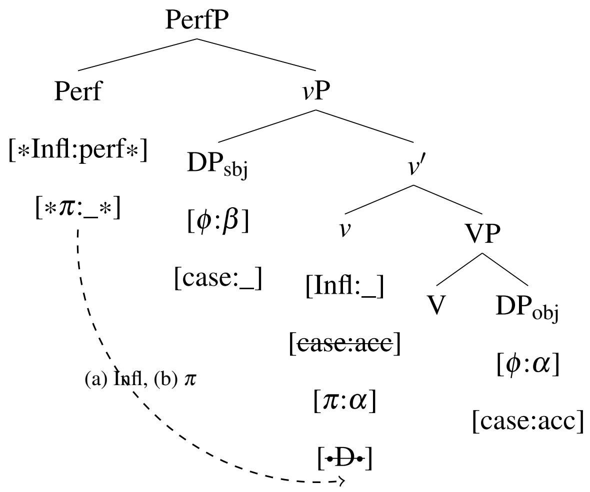
case and agrees with the DP$_{\text{obj}}$ for person (Chomsky 2000, 2001). It also introduces the external argument via the selectional feature [•D•]. These operations are shown in (9).

(9)



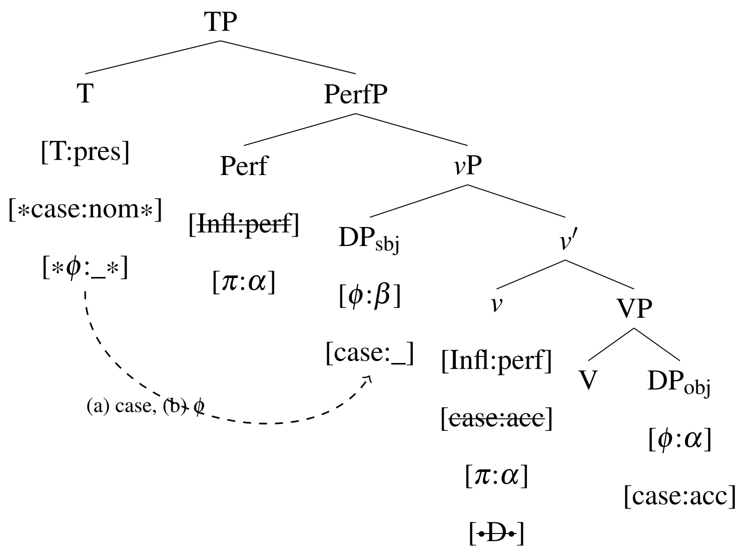
This *v*P is then selected by the head Perf, as illustrated in (10). Recall that the features on Perf are ordered as in (7): [∗Infl:perf∗] $\succ$ [∗π:_∗]. The probe [∗Infl:perf∗] needs to find another Infl-feature to be checked with. The head *v* contains an instance of [Infl:_], it is c-commanded by the probe [∗Infl:perf∗], it is accessible (in accordance with the PIC), and it is the closest matching goal (meaning that there is no other instance of [Infl] that intervenes between the probe on Perf and this potential goal on *v*). Therefore, Perf agrees with *v* for [Infl]. The value *perf* is copied onto *v*. After this probe has been discharged, Perf goes on with the second one, [∗π:_∗]. In the c-command domain of Perf there are two possible matching goals with a [π] value: the DP$_{\text{sbj}}$ and *v* (the DP$_{\text{obj}}$ is invisible because of the PIC). Crucially, the head Perf has already carried out an operation ([∗Infl:perf∗]) before starting this new search ([∗π:_∗]). Nested Agree prescribes that the second probe-feature ([∗π:_∗]) on the probe-head Perf should target the same goal-head as the previous probe-feature ([∗Infl:perf∗]) did, i.e. *v*. Hence, [∗π:_∗] agrees with *v* for [π]. The DP$_{\text{sbj}}$ lies outside the search domain of the probe [∗π:_∗] because Perf is forced to use the open channel already established by the previous probe.

(10)



The derivation proceeds with Merge of T, as shown in tree (11). The T head assigns nominative case to the DP_{sbj} in its c-command domain. Thereafter, it carries out ϕ-Agree. Again, there are two potential goals in the c-command domain: Perf and the DP_{sbj}. Because of the previous operation of case assignment, Nested Agree imposes that the ϕ-probe on T targets the subject in Spec,*v*, instead of Perf. This results in subject agreement on the finite verb.

(11)



The syntactic derivation is now completed. Two complex heads are formed via head movement, V+*v* and Perf+T, given in (12) and (13).

(12)

v

V v

[Infl:perf]

[π:α]

(13)

T

Perf T

[Infl:perf] [T:pres]

[π:α] [ϕ:β]

At Spell-out, the head V in (12) is substituted by the root of the lexical verb, while v is spelled out by participial morphology because of the feature [Infl:perf]. The Vocabulary Items (VIs) in competition for the head Perf in Standard Italian are provided in (14).[11]

(14) *Vocabulary Items for Perf in Standard Italian*

a. /HAVE/ $\leftrightarrow$ Perf[π:α]

b. /BE/ $\leftrightarrow$ Perf elsewhere

The exponent HAVE (14-a) is inserted when the head Perf has agreed for [π] with another head. BE is the default exponent, which is instead inserted when no π-value could be copied onto the probe. Hence, the person probe on the head Perf determines how this head is morphologically realized at Spell-out. Note that the person feature on Perf is not spelled out as person inflection on the verb (as is the case for subject-verb agreement in Italian and English), but rather as allomorph selection. In (13), which corresponds to the derivation of (4-a), the head Perf is substituted by HAVE (in particular, by the specific allomorph /a/, which is chosen in the context of the features [ϕ:-part,+sg] on T) because it bears a valued π value. The T head is substituted by morphological agreement for [ϕ:-part,+sg].

[11] These VIs are specific to the perfect head. Although the problem cannot be fully addressed here, it is conceivable that perfect, possessive, and copula auxiliaries are realized by the same VIs, as suggested by Kayne (1993). Recently, Myler (2014, 2018) has proposed a transitivity-based allomorphy account of possessive and copula auxiliaries. The auxiliary HAVE is the realization of the head v_{BE} in the context of transitive Voice, whereas BE is the elsewhere form. Since transitive Voice involves a π-probe, I believe that Myler's account can be easily translated into an Agree-based allomorphy account, as the one adopted here. The general idea would be that in Standard Italian there is a generic Aux head that is spelled out as HAVE when Agree succeeds, and as BE when it fails. This would be the case of Perf, as indicated by the VIs in (14). Existential and predicative structures may result into BE if they have an unaccusative structure. The possessive HAVE could simply be the effect of a transitive argument structure, while possessives of the type BE + PP can be explained if Agree cannot take place within a PP. More research is needed to explore this connection.

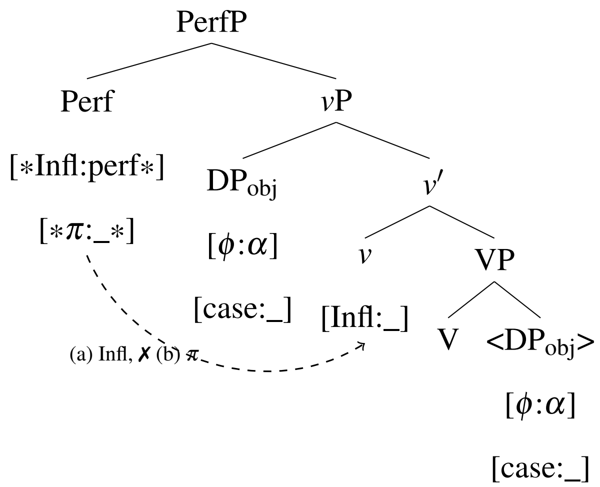
3.4.2 Unaccusative clauses

Unaccusative verbs require the auxiliary BE (cf. (4-c)). In this section, I show that this happens because person Agree on Perf fails. I assume that unaccusative verbal roots are selected by a defective *v* (Chomsky 2000, 2001). This head does not bear a person probe, nor does it assign accusative case, nor does it introduce the external argument, nor does it assign the external Theta-role. In addition, I consider defective *v* to be a phase head, as other *v* heads are (Legate 2003; Richards 2005; Müller 2010, 2011; Abels 2012; Heck 2016). Evidence for this fact comes, for instance, from quantifier floating.

The derivation proceeds as follows. V selects its internal argument and is selected by defective *v*. No features are checked at this point because *v* only bears [Infl:_]. When the head Perf is merged in the structure, it firstly probes for [∗Infl:perf∗]. Agree between Perf and *v* for [Infl] is successful. Subsequently, the second feature on Perf must be discharged. Due to Nested Agree, [$*\pi$:_∗] on Perf also probes *v*. However, in this case *v* does not bear the relevant feature: π-Agree is not successful. Even if Perf can continue its downward search when the goal favored by Nested Agree is not a matching one (downward expansion, see section 2), it cannot find any further goal. In fact, the complement of *v* is opaque for any further syntactic operation because of the PIC. In addition, the DP_{obj} is now located in Spec,*v* (it must move out of the phase domain to remain accessible to the higher case assigner T), which is not an accessible position for the π-probe on Perf according to Nested Agree, unless assuming backtracking.[12] Hence, π-Agree on Perf fails: the person feature on Perf remains unvalued. This is represented in tree (15).

[12] Here I exclude the option of starting a new search from scratch, once the first Agree cycle has failed. Many scholars propose cyclic expansion (Béjar & Rezac 2009) in case the probe has not been satisfied yet after a first cycle of Agree. In principle, nothing prohibits this option. However, if Agree could start again from Perf, Perf would successfully π-agrees with the subject in the second Agree cycle, leading to HAVE insertion with unaccusative verbs. Although cyclic expansion is certainly an option in other languages and for other features, it seems that Standard Italian should not allow for this repair operation in this context (see Murugesan (2022: 64) for an example of parametrization of cyclic expansion). More in general, under the assumption that failed Agree is possible, there is no reason to start a new search when the search domain is exhausted without having found a matching goal. On the contrary, the continuation of the search downwards beyond the Nested-Agree-goal can be considered as part of the first cycle of Agree. Nested Agree only sets up the upper boundary for Agree. However, nothing prevents the probe from keeping searching its c-command domain after the Nested-Agree-goal is inspected as the starting point of the search.

(15)



Because of failed Agree, Perf is substituted by the default allomorph BE, see (14-b) (in particular, in the derivation of (4-c) by /ɛ/, inserted when T bears [ϕ:-part,+sg]).

3.4.3 Reflexive clauses

When a transitive verb is merged with a reflexive clitic pronoun, the perfect auxiliary is BE (cf. (5-a)–(5-b)). Under the present approach, the emergence of the auxiliary BE must be due to lack of valuation of the person probe on Perf (cf. (14)). I propose that this is the direct consequence of the featural defectiveness of the reflexive clitic pronoun (see also Schwarze 1998: 103-104; Alexiadou et al. 2015: 102 for a similar intuition). Reflexive pronouns are referentially defective DPs that enter the derivation with a set of unvalued ϕ-features, which are valued by a c-commanding antecedent DP via binding (Reuland 2001; Fischer 2004; Heinat 2006; Rooryck & Vanden Wyngaerd 2011).[13]

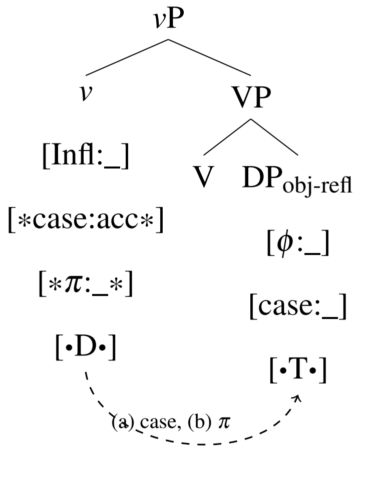
In what follows, I adopt the transitive-unergative analysis of reflexive clauses, which has been proposed by De Alencar & Kelling (2005); Sportiche (2014); Alexiadou et al. (2015), among others. Although auxiliary selection in direct reflexive clauses such as (5-a) would also

[13] I consider binding as a form of coindexation under c-command. Binding consists of valuation of the index on the reflexive by the index on the antecedent (or, following Hicks (2009), of the feature [var:_] on the anaphor by the feature [var:i] on the binder). Consequently, the reflexive pronoun and the c-commanding DP bear the same index and form a referential dependency. Binding is actually independent of ϕ-valuation, which is a consequence of coindexation if one of the involved items carries unvalued ϕ-features. If the reflexive pronoun already bears lexically valued ϕ-features, binding does not result in ϕ-valuation, but only in coindexation. I suggest that this is the case for the reflexive phrase *se stess-o/a/i/e*, which in fact determines HAVE insertion: *Teresa ha lavato se stessa* 'Teresa has(HAVE) washed herself' (see also Reuland 2001 for a similar distinction between *se-reflexive* and *self-reflexive* in terms of valued/unvalued features).

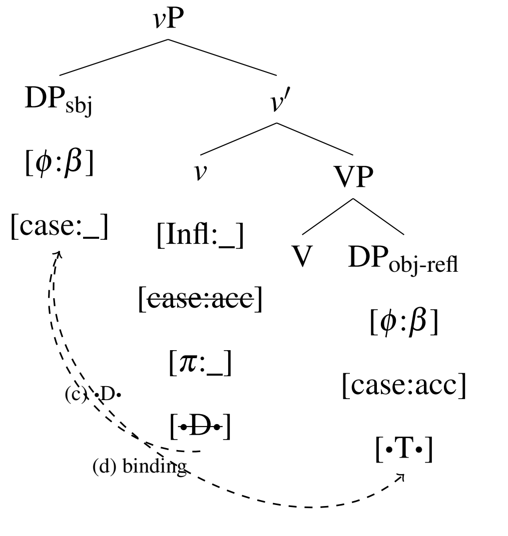
be compatible with the unaccusative analysis of reflexives (see Kayne 1975; Grimshaw 1982; Marantz 1984; Embick 2004, among others), the unaccusative analysis cannot be easily extended to indirect reflexive clauses such as (5-b), which contain an internal argument and an external argument that are clearly distinct.

The derivation of direct reflexive clauses such as (5-a) is similar to the derivation of transitive clauses proposed in section 3.4.1. The only difference here is that the object is a ϕ-defective item. It is also a clitic, which must incorporate into T. I indicate this requirement with a Merge feature [•T•] that must be checked by means of incorporation into T. First of all, *v* assigns accusative case; then, it agrees for person. Recall that Agree is simply the search for a matching feature, independently of valuation. The internal argument contains a matching feature, although unvalued. Therefore, *v* agrees with the $\mathrm{DP}_{\mathrm{obj}}$ for [π], although it cannot copy any π-value, as shown in (16).

(16)



(17)



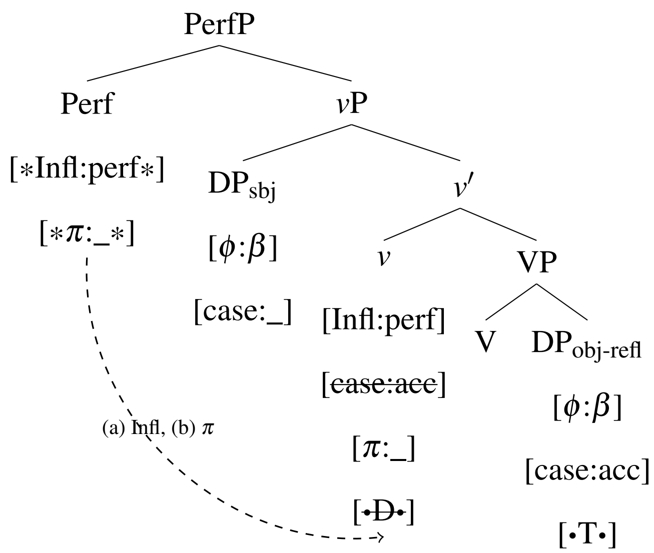
When all operations within the complement of *v* have been carried out, *v* introduces the external argument, as shown in (17). Merge of the external argument, triggered by the feature [•D•], must happen after case assignment and π-Agree because of the *Strict Cycle Condition (SCC)* (Chomsky 1973: 243).[14] As soon as the external argument is introduced in Spec,*v*, binding takes

[14] The *Strict Cycle Condition* states that "no rule can apply to a domain dominated by a cyclic node A in such a way as to affect solely a proper subdomain of A dominated by a node B which is also a cyclic node" (Chomsky 1973: 243). There are different interpretations of this principle depending on what counts as a cyclic node. Under a strict interpretation, every projection (including minimal projections) is a cyclic domain. As observed by McCawley (1984, 1988), if every projection is a cyclic node, the SCC can predict the order of operations (see also Georgi 2014; Müller 2021). In the present case, the alternative feature ordering [•D•] $\succ$ [$*\pi$:_$*$] would not comply with the

place: the two DPs are now coindexed, and the reflexive clitic acquires the same ϕ-features as the DP_{sbj}.[15] Merge of the external argument (and binding) after π-Agree on v results in a relation of counter-feeding: v is in the right structural position to get a person feature from the reflexive object, but this information is provided too late for v. For this reason, the π-feature on v remains unvalued (see also Murugesan (2022) for a similar analysis of the anaphor–agreement effect).

In the next step in (18), Perf selects the vP. Perf checks its [$*$Infl:perf$*$] feature with v. Afterwards, Nested Agree forces the second probe on Perf [$*\pi$:_$*$] to also probe v, exploiting the Agree-Link created by the previous Infl-probe. The person feature on v, although unvalued, satisfies the probe. Thus, just as π-valuation has failed on v in (16), π-valuation on Perf in (18) also fails.

(18)



Given the vocabulary entries in (14), the condition of insertion of the allomorph HAVE is not met,

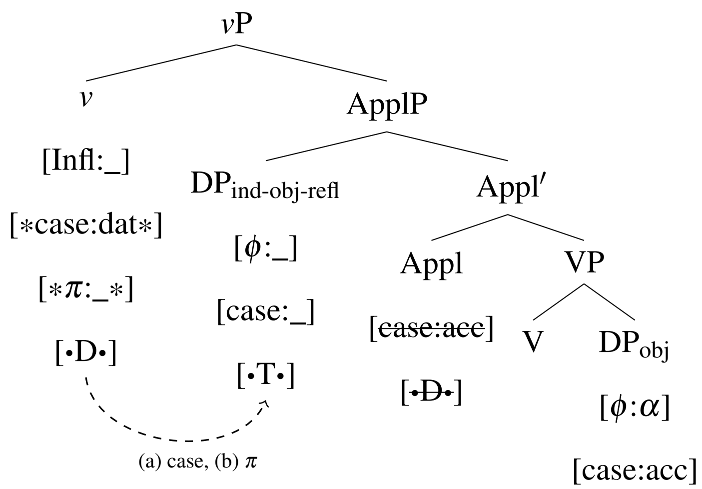
SCC: the earlier discharge of [•D•], which creates the specifier, would make it impossible for v to agree for [π] with the internal argument, since this Agree operation involves the head and its complement with the exclusion of the specifier.

[15] After binding, the reflexive pronoun bears valued ϕ-features: it is spelled out with specific ϕ-features, and can control participle agreement (cf. (5-a)). I analyze Italian past participle agreement as Agree on v for gender and number (*not for person*), which shows up when the object moves out of its base position because it must incorporate into the finite verb or reach the subject position (see section 4.3.2). Participle agreement is the "byproduct" of an edge feature that triggers movement out of the phase domain when the phase is completed (i.e., after binding by the external argument: it is precisely for this reason that participle agreement targets the *valued* ϕ-features of the reflexive clitic). Note that the presence of gender and number features on v due to participle agreement is not relevant for Agree on Perf because Perf probes v for *person*, and not for number or gender. Hence, past participle agreement and auxiliary selection are independent phenomena (see also Belletti 2017). This is also particularly evident in many Italian dialects, where BE is not always associated with participle agreement (Loporcaro 1998).

and the elsewhere form BE is inserted (in particular, for (5-a) the allomorph /ɛ/, corresponding to the features [ϕ:-part,+sg] on T).

The derivation of indirect reflexive clauses such as (5-b) is very similar to the one just sketched. Again, the underlying assumption is that the reflexive clitic pronoun enters the derivation with unvalued ϕ-features. The reflexive pronoun is now merged as an indirect object. I assume that this indirect object is introduced by an applicative head, which brings in the benefactive semantics and assigns accusative case to the DP object (Ura 2000; Anagnostopoulou 2004; Pylkkänen 2008, among others). The applicative phrase is selected by a special type of *v*, called *quirky v*, similar to the transitive one, but assigning dative case (Anagnostopoulou 2001). The derivation is given in (19).

(19)



The head *v* assigns dative case to the reflexive pronoun. Thereafter, it probes for person. The closest matching goal is the reflexive pronoun, which bears the relevant feature [π], although unvalued. The probe targets this goal, even though π-valuation cannot succeed. This is a case of defective intervention (Chomsky 2000: 123): an unvalued matching feature causes the probe to stop its search without valuating it, and prevents it from reaching a lower, valued goal (the DP_{obj}).

After (19), the derivation proceeds in the same way as for direct reflexive clauses. In particular, failed π-valuation on *v* leads to the same result on Perf. At Spell-out, BE is inserted because Perf does not contain any valued π-feature.

The derivation of reflexive clauses has shown that not only does the type of *v* determine the morpho-phonological realization of Perf, but also the π-feature on *v* does, valued in a prior Agree relation between *v* and a lower DP. The person feature of the internal argument is the decisive factor for auxiliary selection.

3.4.4 Impersonal clauses

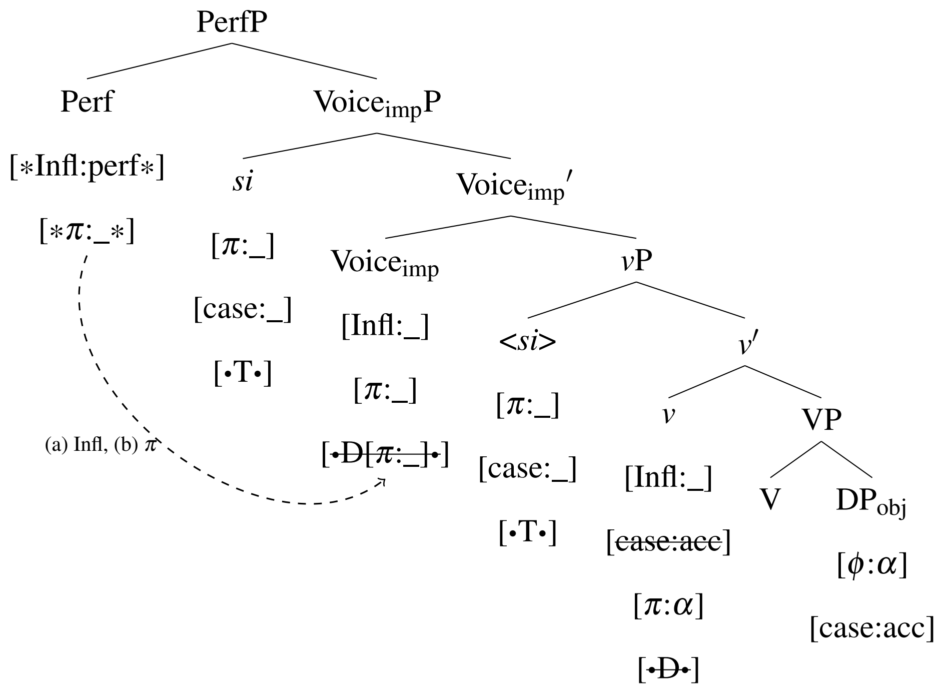
The perfect auxiliary in impersonal clauses is always realized as BE, independently of the type of verb involved (cf. (5-d)). I argue that this is due to an impersonal Voice head, $\text{Voice}_{\text{imp}}$. The auxiliary BE emerges because an unvalued person feature on $\text{Voice}_{\text{imp}}$ always leads to defective intervention: Agree stops without any feature being copied on Perf.

I consider the impersonal pronoun *si* to be a verbal argument (Burzio 1986; D'Alessandro 2004, 2017a). I also assume that it bears a defective set of ϕ-features, in particular an unvalued [π] feature (for similar assumptions, see Burzio 1986; Manzini 1986; Cinque 1988). I suggest that this π-defective pronoun must be licensed by a special impersonal $Voice_{imp}$ head with an unvalued person feature (for comparable proposals, see Legate 2014; Alexiadou et al. 2015). The unvalued π-feature on $\text{Voice}_{\text{imp}}$ licenses the impersonal argument by restricting the specifier of $\text{Voice}_{\text{imp}}$ to arguments that bear an unvalued person feature as well (following again Legate 2014). I express this requirement with the feature [•D[π:_]•]. For this reason, only an impersonal argument (with an unvalued π-feature) can saturate Spec,$\text{Voice}_{\text{imp}}$. The impersonal clitic *si* can be either merged directly in Spec,$\text{Voice}_{\text{imp}}$, or raised to Spec,$\text{Voice}_{\text{imp}}$ from the internal argument position (depending on the argument structure).

In (20), I illustrate the derivation with a transitive *v*P and the impersonal argument base-merged as the subject, but here and below nothing would change with a defective *v*P.[16]

[16]In Standard Italian, transitive impersonal clauses come in two variants: *si=sono mangiat-i spaghetti* vs. *si=è mangiato spaghetti* 'one has eaten spaghetti'. In the former, there is object agreement, past participle agreement, and the object bears nominative case; in the latter, there is no agreement, the verb is in default third person singular, and the object is in accusative case. I propose that the variation between the two versions is a consequence of optionality in the type of *v* that $\text{Voice}_{\text{imp}}$ can select. In the former, the transitive verb combines with a defective *v* (similarly to what happens in passives); in the latter $\text{Voice}_{\text{imp}}$, selects a transitive *v*P (as in (20)); see also Amato (2023).

(20)



After the vP is built, it is selected by Voice$_{\text{imp}}$. The feature [•D[π:_]•] raises the impersonal *si* to Spec,Voice$_{\text{imp}}$. Thereafter, the head Perf enters the derivation, and probes for [Infl]. Voice$_{\text{imp}}$ bears an [Infl] feature, as other v and Voice heads do. Perf finds the [Infl] feature on Voice$_{\text{imp}}$, which acquires the value *perf*. Consequently, the complex head V+v+Voice$_{\text{imp}}$ that arises via head movement will be spelled out as a past participle. The [Infl] feature on v remains unchecked, but this is not problematic: it is exactly what happens in the present tense, when v is selected by T (and not by Perf). The next operation performed by Perf is the search for a person feature. Due to Nested Agree, Perf must probe Voice$_{\text{imp}}$ for [π]. Voice$_{\text{imp}}$ is a matching goal because it contains a π-feature, although unvalued. Therefore, Perf agrees with Voice$_{\text{imp}}$ for person, without copying any value.[17] At Spell-out, the lack of a valued feature on Perf determines the insertion of the elsewhere form BE.

[17]Passive clauses, whose perfect auxiliary is always BE, require a similar analysis. I assume that these clauses contain a passive Voice$_{\text{pass}}$ (Collins 2005; Bruening 2013; Legate 2014; Alexiadou et al. 2015). Perf agrees with Voice$_{\text{pass}}$ for [Infl]. When it goes on to [π], Voice$_{\text{pass}}$ does not contain any ϕ-features. Agree stops because Voice is a phase (or, alternatively, because Voice$_{\text{pass}}$ selects for a defective v with no π-probe). I consider Voice heads as phases because they are types of v. Note that this analysis works independently of the presence or absence of a silent external argument in the passive. As a side note, if impersonal Voice$_{\text{imp}}$ is a phase, the present analysis of impersonals can be simplified in not requiring an unvalued π-feature on Voice$_{\text{imp}}$.

3.4.5 Summary

In argument-structure-driven auxiliary selection, the dependency on both the argument structure and the features of the arguments is modeled via Agree for person between Perf and *v* (or Voice). The featural makeup of both *v* and the arguments determines the presence or absence of a valued person feature on Perf, which is the relevant factor for auxiliary selection. Whenever *v* successfully agrees for person (i.e., in transitive clauses), Perf also bears a valued person feature. In this case, Perf is spelled out by the more specific allomorph HAVE in Standard Italian. Instead, the elsewhere form BE emerges when the head Perf is unable to copy a person value because of morpho-syntactic defectiveness. In unaccusative clauses, π-Agree on Perf fails because *v* is defective (i.e., it does not bear a person probe). In impersonal clauses, the same happens because Voice bears an unvalued person feature. In reflexive clauses, the person probe on Perf cannot be valued because *v* has agreed with a defective DP. The elsewhere form BE emerges whenever there is some kind of featural defectiveness in the structure, in either the functional head *v* or the DPs.

3.5 Auxiliary selection in Italo-Romance

In many Italo-Romance varieties, auxiliary selection is subject-driven: the auxiliary is determined by the person feature of the subject, independently of the argument structure. The most common pattern is BBH (BE with 1st-2nd person subjects, HAVE with 3rd person subjects), but many other distributions are attested (Manzini & Savoia 2005; Loporcaro 2007; Ledgeway 2019). An example of BBH comes from Ariellese (Abruzzese), as shown in (21).

(21) a. Tu si fatijate.
2SG.NOM be.PRS.2SG work.PRTC
'You have worked.'

b. Esse a fatijate.
3SG.F.NOM have.PRS.3SG work.PRTC
'She has worked.' (D'Alessandro & Roberts 2010: 43)

Subject-driven auxiliary selection in Abruzzese has already been analyzed as ϕ-Agree by D'Alessandro (2017b). Nested Agree provides an explanation for the difference between

Abruzzese and Standard Italian: different extrinsic feature orderings. In subject-driven systems, the features on Perf are ordered as in (22) (to be compared with the ordering in (7) assumed for Standard Italian).[18]

(22) $[*\pi:\_*] \succ [*\text{Infl:perf}*]$ (subject-driven systems)

If the features on Perf are ordered as in (22), the person probe always targets the subject, which is the highest matching goal. Hence, the dependency of Perf on the person feature of the subject is realized via Agree with the DP_{sbj}.

At Vocabulary Insertion, the head Perf is realized depending on the result of π-Agree. Different auxiliary distributions can be simply derived with different inventories of vocabulary entries.[19] The VIs for Ariellese are given in (23).

(23) *Vocabulary Items for Perf in Ariellese*

a. /HAVE/ $\leftrightarrow$ Perf[π:-participant]

b. /BE/ $\leftrightarrow$ Perf elsewhere

This analysis predicts that subject-driven auxiliary selection cannot be sensitive to either argument structure or the features of the object because of minimality. This is confirmed by Ariellese reflexive clauses, which exhibit the distribution BBH as transitive clauses do (R. D'Alessandro, p.c.).

Another prediction is that argument-structure-driven systems can show sensitivity to the subject, since in these languages there is subject agreement (i.e., the ϕ-features of the subject are represented on T, to which Perf moves). Indeed, in several varieties the auxiliary depends on

[18] As pointed out by an anonymous reviewer, a substantial question concerns the acquisition of these different orders. I believe that, in the domain of auxiliary selection, children receive enough input to establish the primacy of argument-structure-related relations in the Italian system, and of subject-related relations in the Ariellese system. They translate this evidence into ordering the verbal/functional feature [Infl] before the nominal/lexical feature [π] in Standard Italian, and vice versa in Ariellese. In other words, these orderings mirror the linguistic data and are, therefore, learnable. I expect that children might undergo a period of confused production when the targeted feature ordering is not fully acquired yet, producing some perfect auxiliaries of the Italian type in Ariellese (and eventually vice versa, unless assuming that one order is the default). However, no study has tested this prediction yet, as far as I know. General concerns about the acquisition of extrinsic feature orderings have also been addressed by Georgi (2014: 252-253). I completely share her opinion on this point and refer the reader to her work for discussion.

[19] For instance, in the variety spoken in Capracotta (Molise), the auxiliary distribution is HBB (Manzini & Savoia 2005: II: 708). This is explained with BE as the elsewhere form, plus the following specific VI: /HAVE/ $\leftrightarrow$ Perf[π:+speaker].

both the argument structure and the ϕ-features of the subject. I call languages of this type *mixed systems*. Such an example is Tufillese (Abruzzese): the pattern is BBH in transitive clauses, and BE in unaccusative and reflexive clauses (Manzini & Savoia 2005: II: 690). This distribution is derived with the "Italian" feature ordering in (7) because there is a dependency on the argument structure. The argument-structure-driven status of mixed systems is confirmed by the fact that their reflexive clauses behave as unaccusative clauses, as happens in Standard Italian (and not in Ariellese). The specific person restrictions are due to contextually-sensitive VIs that make reference to the π-feature on T (acquired via Agree with the subject). For instance, Tufillese requires the VIs in (24).

(24) *Vocabulary Items for Tufillese*

a. /HAVE/ $\leftrightarrow$ Perf[π:α] / T[ϕ:3sg]

b. /BE/ $\leftrightarrow$ Perf elsewhere

A final advantage of the Nested-Agree-analysis of auxiliary selection is that closely related varieties have the same syntax. All the cross-linguistic differences are due to the lexicon: different feature orderings on Perf, and different vocabulary inventories. The difference in auxiliary selection between Standard Italian and Italo-Romance varieties is also reduced: not only are they both person-sensitive, but also some alleged subject-driven varieties (mixed systems) are indeed argument-structure-based systems, as Standard Italian is.

4 Further applications of Nested Agree

4.1 Not all apparent minimality violations can be rescued

4.1.1 Case and agreement in ditransitive clauses

Cross-linguistically, some combinations of case and agreement in ditransitive clauses are not possible. In this section, I show that Nested Agree makes the right predictions about cross-linguistic variation in ditransitive clauses. The exclusion of non-attested patterns provides further support for this principle.

Ditransitive clauses contain a recipient/benefactive argument (DP_R) and a patient/theme argument (DP_T). Different patterns of agreement and case marking are attested cross-linguistically. In so-called *indirective case alignment*, DP_T is marked as a transitive object (i.e., with accusative or absolutive case), and DP_R is differently marked (i.e., with dative case). In *secundative case alignment*, the opposite holds: the recipient is marked as a transitive object, whereas the theme has a different marking (i.e., nominative or instrumental). Agreement is subject to the same types of alignment. If a language has only one instance of object marking, this may target either DP_T (*indirective agreement alignment*), or DP_R (*secundative agreement alignment*).

Based on a sample of 124 languages, Bárány (2024) has shown that only three possible combinations of case and agreement alignments are attested, revealing a typological gap: there are no languages with secundative/neutral case and indirective agreement (see Bárány 2024 for a detailed discussion of potential counterexamples). Table 1 illustrates this point.

Table 1: Attested and unattested combinations of alignment types (Bárány 2024: 393).

	Secundative/neutral case	Indirective case
Secundative agreement	✓ (91 languages, e.g. Nez Perce)	✓ (25 languages, e.g. Amharic)
Indirective agreement	✗	✓ (8 languages, e.g. Hungarian)

Nested Agree exactly predicts the typological gap described in Table 1. To illustrate this point, I adopt some standard assumptions about ditransitive clauses (see also Bárány 2021, 2024). Firstly, the head responsible for object agreement is *v*, and it c-commands both DP_R and DP_T. Secondly, DP_R c-commands DP_T (at least in the constructions considered in this sample). Thirdly, ditransitive sentences contain an applicative head, which may assign dative case (this point is subject to cross-linguistic variation). In addition, I propose that the head *v* bears two probes ([$*$case:acc$*$], [$*\phi$:_$*$]), which are extrinsically ordered.[20] The feature ordering is subject to cross-linguistic variation. I also assume that case assignment is not subject to defective intervention: a DP, whose case has already been valued, is invisible for further case assigners.

Before delving into the derivations of the possible ditransitive alignments, let me spend a

[20]The head *v* can also assign dative case, as I have proposed for Italian in section 3.4.3. This does not change the discussion here below. In fact, a quirky *v* assigns dative to DP_R and agrees with it under both feature orderings ([$*$case:dat$*$] $\succ$ [$*\phi$:_$*$], [$*\phi$:_$*$] $\succ$ [$*$case:dat$*$]), resulting in indirective case and secundative agreement.

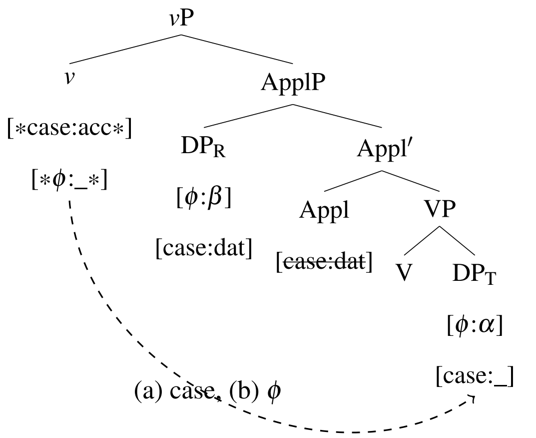
few words on defective intervention, since it plays a crucial role here and in sections 4.1.2 and 4.2.2. This term indicates the scenario where a higher item prevents some lower items from satisfying some feature, although it cannot satisfy this feature itself (Chomsky 2000: 123; see also Richards 2008; Preminger 2014: 130-174 for discussion). In the previous literature, all examples of defective intervention involve either ϕ-Agree or movement, while I am not aware of any example concerning case assignment. A notorious example is agreement in DAT-NOM constructions in Icelandic; note that even in this scenario nominative case assignment by T proceeds without any disruption by the intervening dative argument (see also section 4.1.2). Ultimately, the question of whether there is defective intervention with respect to case assignment is an empirical one. There is empirical evidence for defective intervention for the EPP-feature and ϕ-Agree (although some cases, such as tough-movement and Romance subject-to-subject raising, may find other explanations; see Bruening 2014 and references therein). On the contrary, there are several examples where a DP with valued case does not disrupt the relation between a case assigner and a DP that lacks case: nominative case assignment in Italian and Icelandic quirky subject constructions and VOS structures, long-distance multiple nominative case assignment in Japanese, case assignment in double object constructions. The reason why this happens has yet to be determined with further investigation.

In addition to the lack of empirical evidence, case was not considered as the target of defective intervention for two main reasons: valued case features were thought to be the source of defective intervention for ϕ-Agree by making the DP *inactive*, and case features were considered not to have independent status and to be valued as a by-product of ϕ-Agree (Chomsky 2000). However, there are other approaches that consider case assignment as independent of ϕ-Agree (for instance, Pesetsky & Torrego 2001; Bhatt 2005; Legate 2008; see also configurational approaches such as *dependent case*, Marantz 1991). In this paper, I adopt the view that each operation must be feature-driven, including case assignment, and that this is independent of ϕ-probing. Case assignment is a relation between a functional head and an argument, working similarly to ϕ-Agree. The reason why ϕ-probing and case-probing may be subject to different conditions, including defective intervention, requires further research.[21]

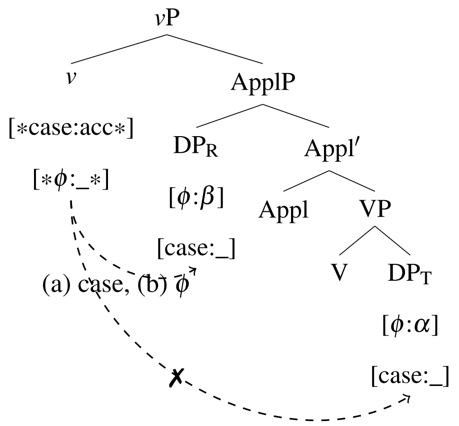
[21] Given the lack of previous literature on defective intervention dealing with case assignment as a probing operation, I can just speculate on the reason why defective intervention may affect ϕ-Agree, but not case assignment.

Coming back now to ditransitive alignments, trees (25)-(26) represent the scenario where case assignment precedes Agree ([∗case:acc∗] ≻ [∗ϕ:_∗]). Two types of languages are possible, depending on case assignment by the applicative head.[22]

(25) *Ind. case + ind. agreement*



(26) *Sec. case + sec. agreement*



In languages where Appl assigns dative (as in tree (25)), DP$_R$ has already been assigned case when *v* discharges its first feature [∗case:acc∗]. Consequently, [∗case:acc∗] on *v* targets DP$_T$, which receives accusative case. This operation creates a dependency between *v* and DP$_T$. As for the second feature [∗ϕ:_∗], Nested Agree prescribes to start its search from the same goal. Hence, *v* agrees with the lower DP$_T$, thereby skipping the higher DP$_R$. This derivation, illustrated in (25), results in indirective case and indirective agreement. Note that it involves an apparent minimality violation (*v* ϕ-agrees with DP$_T$ across DP$_R$) that can be easily explained with Nested Agree.

In other languages, the Appl head does not assign dative case (as in tree (26)). Here, DP$_R$ is a matching goal for the case probe on *v* because it still bears an unchecked case feature when *v* carries out its operations. Hence, *v* assigns accusative to DP$_R$. After this step, Nested Agree

Possibly, this could lie in the difference between valued and unvalued probes. A valued case feature, which is c-commanded by a case assigner and c-commands another unvalued case feature, does not intervene between the assigner and the unvalued feature because the feature responsible for case assignment is a valued probe that looks for a matching *unvalued* feature, unlike unvalued probes, which simply look for a matching feature.

[22]In these trees, dative case assignment takes place in a Spec-Head configuration, as is generally assumed for applicative heads. However, if case assignment is an instance of Agree, then upward case assignment in Spec,Appl is in contradiction with the assumption that Agree only happens downwards, as noted by an anonymous reviewer. I acknowledge this flaw, and suggest that perhaps inherent case assignment may happen upwards, differently from ϕ-probing, which is always downwards. Further work is needed to clarify this point, to establish whether the applications of upward Agree may be reduced to downward Agree (see Bárány & van Der Wal (2022) for a discussion), and to define how Nested Agree deals with upward Agree.

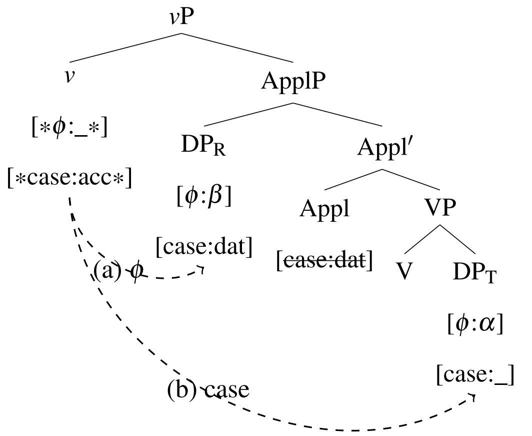
forces the second probe on v to target DP_R. Consequently, v agrees with the higher DP_R, and the lower DP_T cannot be reached anymore, as shown in (26). Languages of this type exhibit secundative case and secundative agreement.

The order of the features on v can also be reversed ([$*\phi$:_$*$] $\succ$ [$*$case:acc$*$]).[23] In this case, only the following scenario is predicted: the ϕ-probe on v must target DP_R because of minimality.[24] In fact, the ϕ-probe on v cannot bypass the higher DP_R because there is no previous dependency on which it could piggyback (as was instead the case in (25)). Hence, under this feature ordering the lower DP_T cannot be ϕ-agreed with, independently of the status of the Appl head as a case assigner. Secundative agreement is the only possibility. As for case, if DP_R already bears dative, it is not a matching goal for case assignment. Consequently, v goes on scanning its c-command domain, finds the unvalued case feature on DP_T, and assigns it accusative case. Languages of this type have indirective case and secundative agreement, as illustrated in (27). Instead, if DP_R does not bear case yet, v assigns it accusative case, resulting in secundative agreement and secundative case, as shown in (28) (cf. (26) for the same outcome; languages of this type are also the majority in the sample under consideration, cf. Table 1).

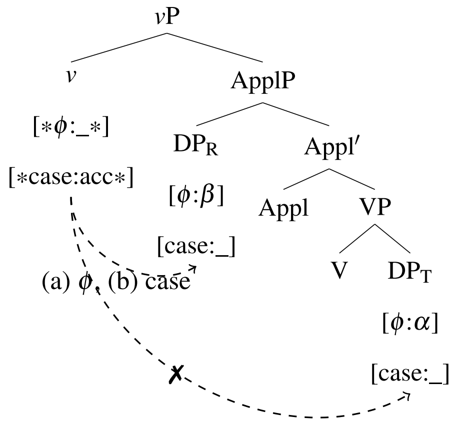
[23] While the primacy of case assignment over ϕ-agreement may be supported by the fact that agreement is case-sensitive in many languages (see section 4.2.2), an anonymous reviewer wonders whether there is evidence for the opposite feature ordering. As illustrated in (27), the order [$*\phi$:_$*$] $\succ$ [$*$case:acc$*$] gives rise to indirective case and secundative agreement. A language of this type is Amharic (see Table 1). Interestingly, it has been noted that in Amharic case assignment and ϕ-agreement do not always go together in other contexts beyond ditransitive clauses, unlike in Indo-European languages (Baker 2012). In particular, object agreement can target nominative items and non-accusative NPs within PPs. Although I cannot provide a full analysis of these data here, it seems to me that they show the independence of ϕ-Agree from case, which may be explained with the order [$*\phi$:_$*$] $\succ$ [$*$case:acc$*$]. Cases where object agreement fails with some accusative arguments (reflexives, wh-phrases, quantifier) can be explained if these do not bear valued ϕ-features, and not as a consequence of the primacy of case assignment.

[24] The accessibility of the dative argument for ϕ-Agree is a further factor for cross-linguistic variation, as empirical evidence shows (see for instance Icelandic in section 4.1.2, where datives cannot be agreed with). According to Bárány (2021, 2024), the different agreement alignments with indirective case in (25) and (27) emerge because languages differ as to whether datives can control agreement (but defective intervention is not considered in his analysis). The approach based on Nested Agree offers the additional possibility of defective intervention: there can be languages where v cannot agree with the dative recipient, but neither can it search beyond it. Moreover, it has the advantage of explaining why there are some languages of type (25) where dative arguments do not control agreement in ditransitives, although they can do so elsewhere (e.g., Ngkolmpu, Basque, Burushaski; Bárány 2024: 407).

(27) *Ind. case + sec. agreement*



(28) **Sec. case + ind. agreement*



Note that, when Appl does not assign case, the effect of the two different feature orderings on *v* is neutralized on the surface, as shown by the same outcome in both (26) and (28) (secundative case and secundative agreement). The crossed-out arrows in (26) and (28) show that in no language *v* assigns case to the higher DP, but agrees with the lower DP. This is ruled out even under Nested Agree and any feature orderings.

More broadly, a generalization emerges from this discussion (and, in general, from all the case studies in section 4). Case assignment can skip DPs that have already been assigned case because they are not matching goals; ϕ-Agree cannot skip any DPs because all DPs have matching ϕ-features; it can do so only if Nested Agree establishes a dependence on a previous operation that creates the relevant configuration.

The derivation of attested and unattested combinations of alignment types in ditransitive clauses provides support for Nested Agree because standard Agree mechanisms would not lead to the same result. In particular, in derivation (25) the higher item $\mathrm{DP_R}$ cannot be otherwise skipped by ϕ-Agree, unless assuming that in some languages dative arguments can be agreed with and in others they cannot. This is the explanation pursued by Bárány (2024). Although case-related accessibility might be a relevant factor even in the present analysis (but see section 4.2.2 for discussion), this cannot be the whole story: there are languages where dative arguments do not control agreement in the configuration in (25), but can do it elsewhere, see footnote 24 (this point is addressed by Bárány (2024: 405-406) by saying that not all datives are alike). Moreover, even though datives might not be targets for Agree in certain languages, it is unclear why they do not lead to defective intervention in (25). If the dative is not skipped by the ϕ-probe, it should intervene at least in some cases. The anti-local character of (25) is not discussed in Bárány (2024). However, ϕ-probing beyond the dative argument should not be possible because of locality. Similarly, a "maximizing" principle such as Maximize Matching Effects (MME) (see appendix 6) would also not help because in (25) the higher DP_R can only satisfy one single probe, whereas the lower DP_T cannot be reached by the ϕ-probe because of minimality (unless MME operates transderivationally and establishes that the two probes should both agree with the object, thereby skipping the recipient). Hence, Nested Agree seems necessary to enable the configuration in (25).

4.1.2 Icelandic DAT-NOM constructions

Nested Agree does not predict that every minimality violation can always be circumvented. In this section, I discuss a case where minimality prevents a probe from bypassing a local goal, independently of the assumed feature ordering.

In Icelandic quirky constructions, the thematic subject bears dative case, whereas number agreement on the verb is controlled by a non-subject nominative argument (Zaenen et al. 1985; Sigurðsson 1996, 1989). Agreement with the lower DP may be blocked by the higher dative, resulting in optionality, as shown in (29-a).[25] Optionality is excluded in biclausal environments, where an intervening dative always blocks agreement, as shown in (29-b).

[25] Optional agreement with the nominative argument in DAT-NOM constructions is the result of speaker variation. Three different varieties of Icelandic have been described: A, B, C (Taraldsen 1995; Holmberg & Hróarsdóttir 2003; Sigurðsson & Holmberg 2008). In DAT-NOM clauses with a preverbal dative, Icelandic A requires number agreement with the nominative object, Icelandic C disallows it, and Icelandic B shows optional agreement. With postverbal datives, as in (29-a), only Icelandic A allows optional agreement (Ussery 2017: 169-170). More generally, the complex factors the regulate agreement in Icelandic include the type of dative argument, the ϕ-features of the nominative argument, the construction, and the dialect (see Holmberg & Hróarsdóttir 2003; Sigurðsson & Holmberg 2008; Ussery 2009, 2017; Kučerová 2016; Murphy 2018, among others). A comprehensive discussion of this variation is beyond the scope of this article. In this section, I will just focus on so-called Icelandic A, where there is optional agreement in simple clauses with postverbal datives as (29-a), but no agreement in biclausal structures as (29-b).

Optionality can be easily accounted for by the analysis proposed in this section, which is based on the availability of two different feature orderings. Whether the two orders are freely available in the grammar (resulting in free variation) or their distribution characterizes various types of Icelandic (resulting in speaker variation), nothing changes for the purpose of the analysis. Indeed, different feature orderings in various subgroups of the speaker community are an appropriate way to model cross-dialectal variation, as pointed out in various parts of this article (e.g., in section 3.5).

(29) a. það líka / líkar sumum stelpunum peningarnir.
3SG like.PL like.SG some.DAT.PL girl.DAT.PL money.NOM.PL.DEF
'Some girls like money.' (Ussery 2009: 2)

b. það virðist / *virðast einhverjum manni hestarnir vera
3SG seem.SG seem.PL some.DAT.SG man.DAT.SG horse.NOM.PL.DEF be.INF
seinir.
slow.NOM
'A man finds the horses slow.' (Holmberg & Hróarsdóttir 2003: 1000)

Nested Agree exactly predicts optional defective intervention in monoclausal sentences (29-a), and obligatory defective intervention in biclausal sentences (29-b). In biclausal structures, Nested Agree does not allow the ϕ-probe to bypass the matching but defective dative goal because there is no previous dependency that can be exploited for this purpose. I will now illustrate why this happens.

I propose that in Icelandic two optional feature orderings on T coexist (see also Sigurðsson & Holmberg 2008 for a comparable analysis): [$*$case:nom$*$] $\succ$ [$*\phi$:_$*$], and [$*\phi$:_$*$] $\succ$ [$*$case:nom$*$]. Monoclausal configurations such as (29-a) have the following syntactic structure: T ... DP_{dat} ... DP_{nom}. Under the feature ordering T[$*$case:nom$*$] $\succ$ [$*\phi$:_$*$], T assigns nominative to the lower DP, assuming that there is no intervention for case assignment (see section 4.1.1 for discussion). Thereafter, Nested Agree forces T to agree with this DP, thereby skipping the dative argument. DP_{nom} controls agreement and there is no defective intervention (i.e., the verb is spelled out as *líka* in (29-a)). Instead, if the features ordering on T is [$*\phi$:_$*$] $\succ$ [$*$case:nom$*$], T agrees with the dative, which bears the relevant ϕ-features, but cannot value the probe (following much of the previous literature on the status of datives as defective goals for ϕ-Agree in some languages: Moravcsik 1974; Rezac 2004; Bobaljik 2008; Preminger 2014; Atlamaz & Baker 2018; Coon & Keine 2020, among others). The ϕ-probe on T remains unvalued, resulting in default agreement (i.e., *líkar* in (29-b)). Thus, optional defective intervention is caused by two possible feature orders.

Biclausal sentences as (29-b) have this structure: T_{fin} ... DP_{dat} ... [$T_{non\text{-}fin}$ ... DP_{nom}]. I assume that in biclausal configurations there is no nominative assignment from matrix T. Nominative on the lower DP is assigned by the lower non-finite T (cf. nominative in non-finite clauses in Italian and Portuguese; Rizzi 1982; Raposo 1987). Alternatively, nominative could come from *v*, as proposed by Sigurðsson (2006) for Icelandic, or be default case (McFadden & Sundaresan 2011). Evidence for nominative assignment within non-finite clauses comes from the facts that objects are marked as nominative also in non-finite clauses that are not part of a biclausal structure (30-a), and that PRO is marked as nominative, as visible on quantifiers (30-b).

(30) a. Að líka svona fáránleiki/*fáránleika!
to like.INF such absurdity.NOM/*ACC
'To like such absurdity!' (Sigurðsson 2006: 293)

b. Strákana langaði til að komast allir í veisluna.
the.boys.ACC wanted for to get all.NOM to the.party
'The boys wanted to all get to the party.' (Sigurðsson 1991: 337)

Assuming that nominative assignment in non-finite clauses happens independently of matrix T, defective intervention in (29-b) can be derived as follows. If the features on matrix T are ordered as [$*$case:nom$*$] $\succ$ [$*\phi$:_$*$], matrix T starts by assigning nominative case. The lower DP already bears nominative case. DPs with already valued case are invisible for case assigners (no defective intervention for case assignment; see section 4.1.1). Thus, this operation fails (see Bošković (2007) for the idea that the case feature on a case assigner head can remain unchecked without causing any problems in the derivation). Since the first probe on T has not found any goal, the second probe [$*\phi$:_$*$] on T is not constrained by Nested Agree: the previous operation has not created any dependency that this probe could exploit. Hence, ϕ-Agree targets the dative argument, which is the highest DP, resulting in default agreement because of defective intervention. The same result is also achieved under the opposite feature ordering ([$*\phi$:_$*$] $\succ$ [$*$case:nom$*$]). T probes the dative for ϕ-features, which acts as a defective intervener: default agreement will be realized on the verb. In addition, nominative assignment by T fails because all arguments already have their case valued.

To sum up, optional agreement across a dative argument in monoclausal configurations can be explained with a flexible feature ordering on T. Optionality breaks down in biclausal contexts, where only default agreement appears, because there is no relevant dependency on which ϕ-Agree could piggyback, independently of the feature ordering. This case study, and in particular the derivation of ϕ-agreement with the nominative object across an intervening dative argument in (29-a), constitutes support for Nested Agree because no other standard Agree mechanism can easily account for it. In fact, if the dative is a defective intervener, only default agreement is expected. If it is not as such, only agreement with it is possible, in accordance with minimality. The internal argument can never be targeted by ϕ-Agree without Nested Agree or without resorting to some specific mechanisms (for instance, one could assume that the probe can continue searching for a goal when it is not satisfied by the dative, but then this is expected to happen without exception in simple clauses and even in biclausal structures).

4.2 Other apparent minimality violations

4.2.1 Person agreement in the perfective aspect in Lak

Nested Agree is a new approach to those Agree dependencies that skip the closest goal. It is a way to account for anti-locality configurations through the derivational timing of operations. In this section, I discuss person agreement in the perfective aspect in Lak, which is controlled by the lowest argument instead of the highest one.

Lak is a Nakh-Daghestanian language. The finite verb exhibits agreement for person (Radkevich & Clemens 2013; Radkevich 2017).[26] In the present tense, person agreement is controlled by the external argument, in accordance with minimality, if this argument is not marked as ergative (31-a). If the external argument bears ergative case, the lower (absolutive) argument controls agreement instead (31-b). Interestingly, in the perfective aspect the finite verb always agrees with the lower argument (31-c)-(31-d). The external argument is skipped even when it is not overtly marked as ergative (31-c).

(31) a. $\boxed{\text{Na}}$ ga Ø-at:a-r-$\boxed{\text{a}}$.
I.1SG I.3SG.ABS I-beat.PRS-1/2.SG
'I usually beat him.'

b. Ga-na-l $\boxed{\text{zu}}$ b-at:a-r-$\boxed{\text{u}}$.
3SG-OS-ERG 2.I.PL.ABS I-beat.PRS-1/2.PL
'He usually beats you.'

[26]Lak has other types of agreement that I do not consider in this discussion (noun class agreement, object agreement, and agreement in biabsolutive clauses). The glosses in (31) are adapted from Radkevich (2017). OS stands for 'oblique stem', I for class I, GER for 'gerund'. 1/2 indicates that markers for first and second person are identical.

c. Na $\boxed{\text{ga}}$ ∅-uwhunu ∅-ur-$\boxed{∅}$.
I.1SG I.3SG.ABS I.SG-ISG.catch.PRF.GER I.SG-AUX-3
'I caught him.'

d. Ga-na-l $\boxed{\text{zu}}$ b-uwhunu b-ur-$\boxed{\text{u}}$.
3SG-OS-ERG 2.I.PL.ABS I.PL-I.PL.catch.PRF.GER I.PL-AUX-1/2PL
'He caught you.' (Radkevich 2017: 4-5)

The anti-local configuration in (31-c) can be explained if person Agree piggybacks on a previous dependency that restricts its search domain. Since this scenario only arises in the perfective aspect, the previous operation must be an inflectional dependency between the perfective aspectual/temporal head and *v*, exactly as in Standard Italian auxiliary selection. The presence of Infl-Agree is indicated by the exponent for the lexical verb, which is a dependent, non-finite verbal form.

The derivation of (31-c) is very similar to the analysis of argument-structure-driven auxiliary selection. The feature ordering on the perfective head must be [$*$Infl$*$] $\succ$ [$*\pi*$]. As illustrated in (32), the perfective head Asp agrees with *v* for [Infl]. Due to Nested Agree, Asp must also probe *v* for person, thereby skipping the external argument. Hence, Asp copies the person feature of the object as it appears on *v*. This feature is realized as inflection on the finite verb (because either T probes Asp, or Asp head-moves to T).

(32) *Person agreement with the internal argument in the perfective aspect*

Asp[$*$Infl:perf$*$],[$*\pi$:_$*$] ... DP$_{\text{sbj}}$[ϕ:α] ... *v*[Infl:_],[π:β] ... DP$_{\text{obj}}$[ϕ:β]

(a) Infl, (b) π-Agree

When the head Asp is not part of the derivation, the lexical verb is not marked for aspect/tense, as in (31-a,b). In this case, there is no Agree for [Infl]. Hence, the person probe on T can freely probe the external argument, in accordance with minimality and case-related accessibility. The highest argument can be skipped in favor of the lowest one only when a previous dependency is established, as in the case of the perfect tense (in both Lak and Standard Italian).[27] Once again,

[27] As highlighted by an anonymous reviewer, this discussion presupposes that perfective aspect requires the presence of a head Asp and Infl-Agree with *v*, whereas non-perfective aspect is morphologically default and does not require Infl-Agree. In Lak, the absence of Asp and Infl-Agree in non-perfective aspect finds a confirmation in the absence of an overt auxiliary and special morphology on the verb in (31-a)-(31-b), which are instead present in the perfective aspect in (31-c)-(31-d). Similarly, in Standard Italian present tense is considered the default, requiring no extra head and no Infl-Agree relation.

Nested Agree can derive the configuration in (31-c), which cannot be explained by standard downward Agree, since ϕ-Agree would always target the higher argument when this is not marked as ergative. Note that a maximize principle like MME could also be used to explain the pattern in (31-c), but it would nonetheless require the assumption of feature ordering because otherwise the ϕ-probe cannot skip the higher argument, in accordance with minimality. Hence, a principle like Nested Agree is needed (see appendix for a discussion of Nested Agree and MME).

4.2.2 Subject agreement in VOS in Spanish

In Spanish, Italian, Catalan, and Romanian, VOS clauses exhibit agreement with the subject across an intervening object (Costa 2000; Cardinaletti 2001; Belletti 2004; Gallego 2013). An example from Spanish is given in (33).

(33) Compraron un libro todos los estudiantes.
buy.PST.3PL a book all the students
'All the students bought a book.' (Gallego 2013: 418)

VOS clauses are analyzed by Ordóñez (1998); Gallego (2013) as the result of object shift (plus V-to-T movement).[28] Because of object shift, the object c-commands the subject. Evidence comes from binding (Gallego 2013: 416) and Principle C violations (Cardinaletti 2001: 129). If the object c-commands the subject, it should intervene for case assignment and agreement between T and the subject.

Gallego (2013) has used *equidistance* to account for these data. Even though equidistance might clarify why the subject is accessible to T, it does not explain why T agrees with the subject and not with the object. An alternative approach has been proposed by Heck (2016: 53-58), but it requires the mechanism of a *separate workspace*. Other accounts link the fact that in languages such as Italian and Spanish agreement on T is always controlled by the nominative argument to

[28] An alternative analysis of VOS structures is based on clause-internal VP-topicalization (see Zubizarreta 1998 for Spanish, Belletti 2004 for Italian). Under this VP-fronting analysis, the object does not c-command the subject out of the fronted VP. Even though the object should not be an intervener for the subject because of lack of c-command, it is still c-commanded by T. Since the object is in the c-command domain of T, it might be the closer possible matching goal for ϕ-Agree on T, depending on the Agree algorithm. For instance, in the smuggling analysis of passives by Collins (2005) a DP within a fronted VP (i.e., the passivized argument) is closer to a probe than a lower DP.

the case-discriminating nature of ϕ-agreement: a ϕ-probe can distinguish between its possible targets based on their case marking, eventually leading to failed Agree, or to ungrammaticality (see Bobaljik 2008; Preminger 2014, among others). However, the dependence of ϕ-Agree on case assignment requires an explanation. Bobaljik (2008) has proposed that this happens because Agree is post-syntactic. Nested Agree offers an alternative view: ϕ-agreement is case-discriminating because features are ordered (as implicitly assumed by those accounts that establish the priority of case and the parasitism of ϕ-Agree). In particular, nominative assignment from T precedes the search of T for ϕ-features ([$*$case:nom$*$] $\succ$ [$*\phi$:_$*$]). The application of Nested Agree guarantees subject agreement, as represented in (34).

(34) *Subject agreement in VOS structures*

T[$*$case:nom$*$],[$*\phi$:_$*$] ... DP_{obj}[case:acc],[ϕ:α] ... DP_{sbj}[case:_],[ϕ:β]

(a) case, (b) ϕ-Agree

In (34), T assigns nominative case to the subject. Although the object is higher in the structure, it is not a possible matching goal for the probe [$*$case:nom$*$], since it already bears accusative case (no defective intervention for case assignment, see section 4.1.1). After T has interacted with the DP_{sbj} for case, it must also agree with it for [ϕ], as prescribed by Nested Agree. Thus, in VOS clauses there is no intervention by the object if case assignment precedes ϕ-Agree.

4.2.3 Bulgarian multiple wh-fronting

When there are two phrases that could move because they both bear the relevant feature, minimality prescribes that only the structurally highest one moves. For instance, in English only the highest wh-phrase (i.e., the subject) moves to Spec,C, when there is more than one wh-phrase. Unlike English, some languages allow for movement of multiple wh-phrases to Spec,C. Interestingly, in some languages such as Bulgarian and Romanian the order of the wh-phrases in the left periphery is exactly the same as the base-Merge order (wh-sbj > wh-obj). An example from Bulgarian is given in (35).

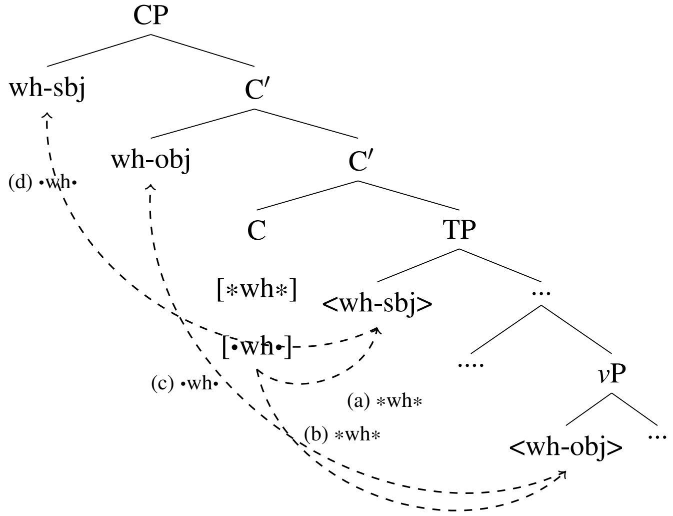
(35) Koj kogo vižda
who.NOM.3SG who.ACC.3SG see.PRS.3SG
'Who sees whom?' (Rudin 1988: 449)

The order of the wh-phrases in (35) is unexpected. Starting from a syntactic structure where wh-sbj is higher than wh-obj, the expected linear order in the left periphery is wh-obj > wh-sbj. In fact, C should first attract the higher wh-sbj, in accordance with minimality, and then the lower wh-obj.

Order-preserving movement of the type in (35) can be justified with Nested Agree. Under the assumption that movement consists of both an Agree operation and a Merge operation (*feature-based* account of movement, following Chomsky 1995b; Abels 2012, among others), the C head carries both an Agree feature and a Merge feature (respectively, [∗wh∗] and [•wh•]). I propose that Agree features are discharged before Merge features: [∗wh∗] ≻ [•wh•]. In addition, I suggest that in languages that allow for multiple fronting (as Bulgarian) movement is triggered by a single feature that can be used multiple times (in this case, by a single pair of [∗wh∗] ≻ [•wh•]). The Agree feature undergoes Multiple Agree, and the Merge feature attracts multiple items (for features triggering Multiple Agree and Multiple Merge, see Hiraiwa 2001; Nevins 2011, among others). The only difference between Bulgarian and English is that in the former the wh-features on C can trigger Multiple Agree/Merge, while in the latter they cannot.

The derivation for Bulgarian is represented in (36).

(36) *Multiple wh-fronting in Bulgarian with Nested Agree*



In (36), C bears the ordered features [$*$wh$*$] $\succ$ [•wh•]. The wh-probe [$*$wh$*$] first targets wh-sbj in Spec,T. Then, it also reaches wh-obj in Spec,*v*, undergoing Multiple Agree. Once the domain of this operation is exhausted, the last goal of [$*$wh$*$] is wh-obj in Spec,*v*. Now C discharges the Merge feature [•wh•]. Nested Agree forces [•wh•] to target wh-obj because it is the last probed element. Therefore, [•wh•] raises wh-obj to Spec,C. The last goal of [•wh•] is now located in Spec,C (the landing site of wh-obj). This position is higher than the one of the probe [•wh•], which is located on C. In this case, I suggest that [•wh•] starts its search again from its position on C because of the c-command requirement. [•wh•] finds wh-sbj in Spec,T and raises it to an outer specifier. This process is repeated until no more relevant goals are found in the c-command domain of C. The resulting surface order is *wh-sbj wh-obj V*, as in (35).

To sum up, when applied to Merge features, Nested Agree has the following two effects: it forces [•wh•] to search into a restricted domain after [$*$wh$*$] has returned a goal lower than C, and it again enlarges the domain for the second application of [•wh•] after its first application has returned a position higher than C (the specifier of the operation-triggering head). This procedure causes the lowest item in the c-command domain of the attracting head to move first, leading to the surface order found in Bulgarian.

4.3 Potential counterexamples

4.3.1 Agreement with unmarked DPs in Hindi-Urdu

In most of the configurations explored so far, case assignment and ϕ-agreement target the same item (see, for instance, section 4.2.2). There are also scenarios where a head assigns case to an argument but ϕ-agrees with another one. We saw such an example in section 4.1.1 (indirective case and secundative agreement, in (27)). Another example comes from Hindi-Urdu, if ergative case is assigned by *v* (as is often assumed): ϕ-agreement with the internal argument can show up with ergative case on the external argument, as illustrated in (37).

(37) raam-ne roṭii khaayii thii
Ram.M-ERG bread.F eat.PERF.F be.PST.F
'Ram had eaten bread.' (Mahajan 1990: 73)

If case assignment and ϕ-agreement result from two Agree operations from *v*, these should be subject to Nested Agree, and should target the same argument. However, sentence (37) shows that these two operations are independent. In this section, I suggest several ways to explain why case and ϕ-agreement may end up targeting different elements, without Nested Agree forcing them to apply together.

First of all, the two responsible features may be interleaved by another feature that destroys the c-command condition of Agree by moving the DP in question out of the way. In this case, the pattern in (37) is derived if the feature ordering on *v* is [$*\phi*$] $\succ$ [•D•] $\succ$ [$*$case:erg$*$] (where [•D•] is the feature responsible for the introduction of the internal argument). The head *v* agrees with the object; then, it introduces the external argument, and it assigns ergative case to it. Note that the precedence of ϕ-Agree over the structure-building feature [•D•] is enforced by the SCC (see also footnote 14). The precedence of [•D•] over case assignment is confirmed by the fact that ergative case is not assigned to the internal argument. Hence, the data in (37) are explained by Nested Agree and the only plausible feature ordering.

An alternative reason for the independence of ϕ-agreement and case assignment is that the relevant features are located on separate heads. I will now offer an alternative analysis of (37) based on this option, which has the further advantage of clarifying why DPs with assigned case are unavailable for agreement. In fact, in Hindi-Urdu ϕ-agreement is always controlled by the higher DP with unmarked case (see Bobaljik 2008 and references therein). In (37), the verb agrees with the object because it is not marked for case, while the higher subject bears ergative case. In (38-a), the verb exhibits default masculine because both arguments are case-marked. In (38-b), the verb agrees with the subject, which is not marked for case, independently of the case marking on the object (which here is unmarked, but could also bear accusative).

(38) a. bacconn̥-ne siitaa-ko dekhaa thaa
children.M.PL-ERG sita.F-ACC see.PERF.M.SG be.PST.M.SG
'The children had seen Sita.'

b. siitaa kelaa khaatii thii
sita.F banana.M eat.IMPF.F be.PAST.F
'Sita (habitually) ate bananas.' (Mahajan 1990: 72-73)

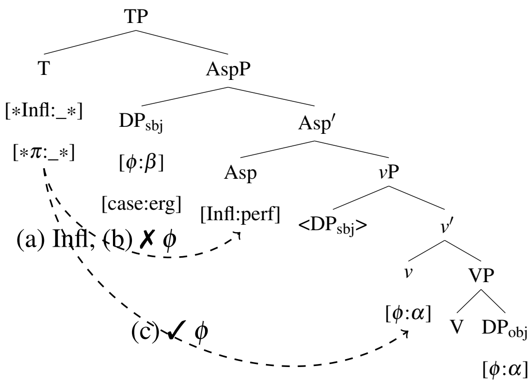
The analysis I am about to present closely resembles the account in terms of anti-locality proposed by Fritzsche (to appear).[29] Four main ingredients are required. Firstly, ergative case comes from a head Asp related to the perfective because ergative case only shows up in the perfective (Bjorkman 2018). Secondly, in Hindi-Urdu DPs must move to obtain overt case marking. Ergative DPs move to Spec,Asp. Specific/animate DPs also move to receive accusative case, which is a type of DOM marking (see Bhatt & Anagnostopoulou 1996; Kalin 2014, among others). Accusative DPs move (at least) to Spec,Asp (cf. Borer 2005; Svenonius 2006; Vainikka & Brattico 2014 for accusative case from Asp). Thirdly, ϕ-agreement is located in T (Mahajan 1989; Anand & Nevins 2006). Fourthly, I assume that T agrees for [Inf] with Asp, and, crucially, Infl-Agree precedes ϕ-Agree. The Infl-probe on T is unvalued. A valued [Infl] feature is located on the aspectual head (as assumed for Lak and Italian in this paper); this also agrees with v, which is spelled out as a participle (this part of the derivation is omitted for the sake of simplicity). With all these ingredients in place, we can explain why ϕ-Agree skips all the case-marked DPs, thereby also deriving the anti-locality configuration in (37) (where ϕ-agreement is controlled by the lower argument), without introducing any anti-locality constraint.

The derivation of object agreement in (37) is given in (39). The subject moves to Spec,Asp to get ergative case. When T enters the derivation, it agrees with Asp for [Infl]. Then, it probes for ϕ. Due to Nested Agree, it must start its search from Asp, thereby skipping the DP_{sbj} in Spec,Asp. Since Asp does not contain ϕ-features, T keeps searching downwards, encountering the ϕ-features on v, which has agreed with the unmarked DP_{obj}.[30]

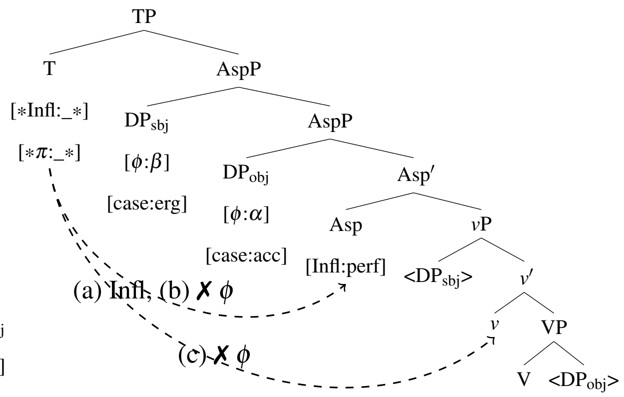
[29] In particular, my first three assumptions are fully shared with Fritzsche's analysis. I add the assumptions of [Infl] on T and feature ordering, while Fritzsche (to appear) introduces an anti-locality constraint on Agree operations.

[30] ϕ-Agree on v is in complementary distribution with case assignment: v probes the internal argument only when this does not bear case. This fact can be modeled if v bears a [DOM] feature that triggers movement of the DP out of its base position. This DOM-probe is discharged first. If there is a [DOM] feature on the internal argument, this is moved. The following probe [$*\phi*$] fails, assuming that ϕ-Agree only happens under c-command (and the SCC would also forbid this Agree operation). If the DP does not bear any DOM-feature, the DOM-probe fails. The subsequent ϕ-probe is free and agrees with the in situ object.

(39)



(40)



Instead, when both DPs are case-marked, both DPs move to Spec,Asp: the subject is assigned ergative case, and the object accusative case. The derivation is illustrated in (40). T agrees with Asp for [Infl]. Then, it probes for ϕ. Nested Agree forces it to start its search from Asp. There is no matching feature in its c-command domain because the two DPs have moved, Asp does not bear ϕ-features, and *v* has also not agreed with the object because this has moved out of the complement position (see footnote 30). ϕ-Agree on T fails, resulting in default agreement, as in (38-a).

Another possibility is subject agreement when the subject is not marked for case, as in (38-b). In this derivation, the $\mathrm{DP_{sbj}}$ does not move. After Infl-Agree with Asp (marked as [Infl:impf]), the ϕ-probe on T agrees with the $\mathrm{DP_{sbj}}$ in Spec,*v*, resulting in subject agreement.[31]

To sum up, potential counterexamples where two Agree operations behave independently can be addressed in various ways. So far, I have proposed some possible solutions: (i) the two probes are separated by a Merge feature, and (ii) the probes are located on different heads. In the next section, I explore a case where other mechanisms are at play.

[31] Unlike Lak (section 4.2.1), imperfective aspect in Hindi-Urdu seems to require a dedicated head and Infl-Agree because it contains an auxiliary and special inflection on the verb. Bobaljik (2008: 309) also reports an example in future tense, originally given by Mohanan (1994: 59). Future tense seems to involve a TAM-projection as well because it contains a dedicated morpheme that used to be a tense auxiliary (Butt & Rizvi 2010). Hence, all tense and aspect types might be treated as involving an aspectual head, as in the above trees (note also that in Hindi-Urdu present morphology does not exist, except for the verb BE; Butt & Rizvi 2010). To propose an accurate analysis of Hindi-Urdu tense/aspect morphology, more clausal types and more syntactic tests should be considered. I leave this topic to future research.

4.3.2 Past participle agreement

Another scenario where case and ϕ-agreement may be dissociated is when the head *v* seems to agree upwards with a moved item in a Spec-Head configuration, but it assigns case to the internal argument (i.e., the mirror scenario to the one discussed in section 4.3.1). In section 4.3.1, I have proposed that case assignment and ϕ-agreement are independent processes if either they involve different heads, or their order bleeds the application of the two operations to the same item. Here I introduce a third possibility: the two operations are independent because they belong to two different stages of the derivation, although the relevant features are located on the same head. I will illustrate this point with past participle agreement (ppa) in Standard Italian.

In Standard Italian, ppa consists of overt morphological inflection for gender and number on the perfect participle. This is controlled by clitic pronouns (canonical, reflexive, impersonal), unaccusative or passive subjects, and nominative objects of impersonal and quirky clauses. For example, in (41) gender and number agreement on the participle is controlled by the object clitic.

(41) Teresa li=ha fatt-i.
Teresa ACC.3PL.M=have.PRS.3SG make.PRTC-PL.M
'Teresa has made them.'

Adopting the standard assumption that ppa is located on *v*, which also assigns accusative case to the internal argument, in principle it can happen that ppa and case assignment do not target the same DP. However, this possibility does not arise in Italian because case assignment and ppa are never dissociated. When there are multiple moving DPs, it is always the lower accusative item that controls ppa; when the controller of ppa is a dative reflexive clitic, *v* assigns dative (see the derivation in (19)); when *v* is unaccusative, it does not assign case. Nonetheless, similar phenomena in other languages point to this possibility more clearly. A comparable phenomenon is number agreement with a moving DP in Dinka (van Urk 2015: 135, van Urk 2020: 122-123). Although this type of ϕ-agreement is considered to be realized in C as the effect of a composite probe that moves the DP and agrees with it, intermediate phase heads bear the same ϕ-probe (van Urk 2015: 137-138). Hence, *v* ϕ-agrees with a moving DP, but eventually assigns case to another DP. In particular, this type of agreement also appears in long-distance movement on all the crossed phasal heads, showing that it is not related to case assignment at all. A similar example is participle agreement in Passamaquoddy: Ā-extraction across a verb optionally triggers participial agreement on it. Bruening (2001: 209) argues that this type of participle agreement is parasitic agreement resulting from movement to the edge of the phase *v*. This type of ϕ-agreement is independent of case assignment and is only possible with movement of the controller.

Similarly to what has been proposed by Bruening (2001); van Urk (2015), I suggest that Italian ppa is a reflex of movement of the controller DP to the edge of the phase head *v*. I follow the well-established idea that ppa is dependent on movement (Burzio 1986; Kayne 1989; Belletti 2017, among others), rephrasing Spec-Head-agreement into Minimalim. Indeed, in Italian all elements that control ppa must move.[32] These moving items bear an unchecked A-feature in the initial part of the derivation: [case:_], or [•T•] (i.e., the feature responsible for the incorporation of clitics into T). Movement is triggered by an *edge feature* (EF) (Chomsky 2001; Müller 2010; Abels 2012; Nunes 2019). The EF is inserted on the phase head *v* when its complement still contains an unchecked A-feature, after all operations on *v* have already been performed.

I propose that in Italian edge features contain three parts: an Agree component [∗F∗] that looks for the A-feature to be moved out of the phase domain, a probe for gender and number [∗γ:_∗, ∗#:_∗] that agrees with the head bearing the unchecked A-feature, and a Merge component [•F•] that moves the individuated head to the edge of the phase. The composition of the EF is illustrated in (42).[33]

[32] Amato (2023) argues that ppa without overt movement (e.g., in unaccusative clauses with postverbal subjects) involves covert A-movement with Spell-out of the lower copy (see also Cardinaletti 1997; Amaechi & Georgi 2020). The DP that controls ppa always moves at least up to Spec,*v*, as shown by the following tests: control into adjuncts, anaphor binding, quantifier scope interactions, quantifier floating, short scrambling.

[33] The single components of the edge feature are also subject to Nested Agree. In particular, Nested Agree is necessary to explain the case of participle agreement with multiple clitics. When there are two possible controllers of ppa in the same clause, ppa tracks the features of the element whose base position is lower, skipping the more local item, as shown in (i).

(i) a. Teresa se=li=è mangiat-i.
Teresa REFL.DAT.3SG.F=ACC.3PL.M=be.PRS.3SG eat.PRTC-PL.M
'Teresa has eaten them for/by herself.'

b. *Teresa se=li=è mangiat-a.
Teresa REFL.DAT.3SG.F=ACC.3PL.M=be.PRS.3SG eat.PRTC-SG.F
'Teresa has eaten them for/by herself.'

Nested Agree provides a simple explanation for the data in (i). Assuming that a single EF undergoes Multiple Agree, the domain of the γ/#-probe on the EF is restricted to the lower clitic by the previous Agree-component of the EF ([∗•T•∗] in this case), similarly to what has been shown for multiple wh-movement in Bulgarian in section 4.2.3.

(42) EF: $[*F*] \succ [*\#*], [*\gamma*] \succ [\bullet F\bullet]$

Ppa is the result of the gender/number-probe in (42). It is the byproduct of downward Agree of an edge feature that triggers successive cyclic movement. Hence, this type of agreement between *v* and a higher element is not a counterexample to Nested Agree because it has a special source (which comes after π-Agree and case assignment by *v*). In this scenario, #/γ-agreement and case assignment are independent because the former is not due to probing of an inherent feature, but to an edge feature. Given its source, it is only carried out after the phase domain is completed (i.e., after case assignment). Only at this point is the edge feature inserted. The two operations belong to two different derivational stages; therefore, they are not related by Nested Agree.

5 Conclusion

In this paper, I have introduced a new tool for dealing with minimality violations, *Nested Agree*. This principle states that the dependency between two heads created by a feature must be exploited by other features on the same heads that are subsequently discharged. Assuming that features on the same head are ordered, the operation that goes first influences the domain of the subsequent operation(s) because its goal must be inspected first, thereby skipping higher goals. Under this perspective, minimality violations do not arise (despite superficial appearance) if the higher, alleged intervener actually lies outside the search domain of the probe. With the formulation of this principle, this study contributes to the ongoing discussion on the conditions of the operations Agree and Merge for multiple probes, in particular on the locality condition, with a focus on intervention effects.

The application of Nested Agree was exemplified in the domain of auxiliary selection in Standard Italian. Nested Agree enables a derivational, explicit analysis of the distribution of the perfect auxiliaries in Standard Italian, which can be easily extended to those languages where auxiliary selection is subject-driven. No previous account can provide such a comprehensive explanation, thus highlighting the need for a principle such as Nested Agree. This principle is necessary to account for argument-structure-driven auxiliary selection because it allows the probe on the perfect head to get past the subject of unaccusative verbs (causing BE insertion) and transitive verbs (leading to BE insertion in reflexive clauses). Note that the analysis presented here also extends to the complicated distribution of perfect auxiliaries in Italian restructuring, which finds an explanation based on Nested Agree (Amato 2024).

Nested Agree was also shown to play a crucial role in many other scenarios. For instance, it makes the right predictions about cross-linguistic variation in ditransitive clauses, and dative intervention in Icelandic, independently of the assumed feature ordering. The exclusion of non-attested patterns and the prediction of cases where intervention cannot be avoided provide further support for this principle. Moreover, it can also be used to solve other cases of apparent minimality violations, such as subject agreement in VOS structures, agreement with absolutive arguments, and multiple wh-fronting with order-preserving movement. New studies, in which problematic configurations for locality can be solved by Nested Agree, will contribute to the discussion of its empirical coverage and predictions, also in relation to other mechanisms already proposed in the literature (see appendix 6 for an initial discussion of this aspect).

Acknowledgments

My greatest thanks go to Fabian Heck and Gereon Müller for helping me develop Nested Agree. I also thank Anna Cardinaletti for her feedback. I am grateful to the whole Institut für Linguistik at Universität Leipzig, especially Barbara Stiebels, Philipp Weisser, Anke Himmelreich, and Rosa Fritzsche. I thank the anonymous NLLT reviewers and the handling editor who provided me with very helpful and constructive comments. I also thank other anonymous readers who commented on earlier versions of this article. I thank the Deutsche Forschungsgemeinschaft for funding. This project is a result of the DFG Graduiertenkolleg Interaction of Grammatical Building Blocks IGRA (GRK 2011), Project number 242675213. Open Access funding was enabled and organized by Projekt DEAL.

References

Abels, Klaus. 2012. Phases: An essay on cyclicity in syntax. Berlin: Walter de Gruyter.

Alexiadou, Artemis, Elena Anagnostopoulou, & Florian Schäfer. 2015. External arguments in transitivity alternations: A layering approach. Oxford: Oxford University Press.

Amaechi, Mary, & Doreen Georgi. 2020. On optional wh-/focus fronting in Igbo: A SYN-SEM-PHON interaction. Zeitschrift für Sprachwissenschaft 39:299–327.

Amato, Irene. 2022. Auxiliary selection is Agree: person-driven and argument-structure-based splits. In Rllt17, ed. Ora Matushansky, Laurent Roussarie, Michela Russo, Elena Soare, & Sophie Wauquier, Special issue of Isogloss Open Journal of Romance Linguistics 8(2)/10, 1–20.

Amato, Irene. 2023. Auxiliary selection in Italo-Romance: A Nested-Agree approach. Linguistik Aktuell/Linguistics Today 281. Amsterdam/Philadelphia: John Benjamins Publishing Company.

Amato, Irene. 2024. Auxiliary selection in Italian restructuring: An insight into the size of the clause. Syntax .

Anagnostopoulou, Elena. 2001. Two classes of double object verbs: the role of zero morphology. In Progress in grammar. Articles at the 20th anniversary of the Comparison of Grammatical Models Group in Tilburg, ed. Marc van Oostendorp & Elena Anagnostopoulou, 1–27. Amsterdam/Utrecht/Delft: Meertens Institute Electronic Publications in Linguistics.

Anagnostopoulou, Elena. 2004. On clitics, feature movement and double object alternation. In Minimality effects in syntax, ed. Arthur Stepanov, Gisbert Fanselow, & Ralf Vogel, 15–36. Berlin: De Gruyter Mouton.

Anand, Pranav, & Andrew Nevins. 2006. The locus of ergative case assignment: Evidence from scope. In Ergativity: Emerging issues, ed. Alana Johns, Diane Massam, & Juvenal Ndayiragije, 3–25. Dordrecht: Springer.

Arregi, Karlos, & Andrew Nevins. 2012. Morphotactics: Basque auxiliaries and the structure of Spell-out. Dordrecht: Springer.

Asarina, Alevtina. 2011. Case in Uyghur and beyond. Doctoral dissertation, Massachusetts Institute of Technology.

Assmann, Anke, Doreen Georgi, Fabian Heck, Gereon Müller, & Philipp Weisser. 2015. Ergatives move too early: On an instance of opacity in syntax. Syntax 18:343–387.

Atlamaz, Ümit, & Mark Baker. 2018. On partial agreement and oblique case. Syntax 21:195–237.

Baker, Mark. 2012. On the relationship of object agreement and accusative case: Evidence from Amharic. Linguistic Inquiry 43:255–274.

Baker, Mark, & Jonathan Bobaljik. 2017. On inherent and dependent theories of ergative case. In The Oxford handbook of ergativity, ed. Jessica Coon, Diane Massam, & Lisa Demena Travis, 111–134. Oxford: Oxford University Press.

Baker, Mark, & Livia Camargo Souza. 2020. Agree without agreement: Switch-reference and reflexive voice in two Panoan languages. Natural Language & Linguistic Theory 38:1–62.

Bárány, András. 2021. A typological gap in ditransitive constructions: no secundative case and indirective agreement. In Proceedings of the 38th West Coast Conference on Formal Linguistics, ed. Una Y. Chow Rachel Soo & Sander Nederveen, 43–53. Somerville, MA: Cascadilla Proceedings Project.

Bárány, András. 2024. Case and agreement alignment in ditransitive constructions: A typological gap and its explanation. Language 100:385–432.

Bárány, András, & Jenneke van Der Wal. 2022. We don't Agree (only) upward. Linguistic Inquiry 53:501–521.

Bech, Gunnar. 1955. Studien über das deutsche Verbum infinitum. Tübingen: Niemeyer.

Béjar, Susana, & Milan Rezac. 2003. Person licensing and the derivation of PCC effects. In Romance linguistics: Theory and acquisition, ed. Ana-Teresa Pérez-Leroux & Yves Roberge, 49–62. Amsterdam: John Benjamins Publishing Company.

Béjar, Susana, & Milan Rezac. 2009. Cyclic Agree. Linguistic Inquiry 40:35–73.

Belletti, Adriana. 2004. Aspects of the low IP area. In The structure of CP and IP. The cartography of syntactic structures, ed. Luigi Rizzi, 16–51. New York: Oxford University Press.

Belletti, Adriana. 2017. (Past) participle agreement. In The Wiley Blackwell companion to syntax, ed. Martin Everaert & Henk C. van Riemsdijk, 2973–3000. Oxford: Blackwell.

Bhatt, Rajesh. 2005. Long distance agreement in Hindi-Urdu. Natural Language & Linguistic Theory 23:757–807.

Bhatt, Rajesh, & Elena Anagnostopoulou. 1996. Object shift and specificity: Evidence from ko-phrases in Hindi. In Proceedings of CLS 32, ed. Lise M. Dobrin, Kora Singer, & Lisa McNair, 11–22. Chicago: Chicago Linguistic Society.

Bjorkman, Bronwyn M. 2011. Be-ing default: The morphosyntax of auxiliaries. Doctoral dissertation, Massachusetts Institute of Technology.

Bjorkman, Bronwyn M. 2018. Ergative as perfective oblique. Syntax 21:321–361.

Bjorkman, Bronwyn Moore, & Hedde Zeijlstra. 2019. Checking up on (ϕ-)Agree. Linguistic Inquiry 50:527–569.

Bobaljik, Jonathan D. 2008. Where's phi? Agreement as a post-syntactic operation. In Phi theory: Phi-features across interfaces and modules, ed. Daniel Harbour, David Adger, & Susana Béjar, 295–328. Oxford: Oxford University Press.

Borer, Hagit. 2005. Structuring sense: An exo-skeletal trilogy. Oxford: Oxford University Press.

Bošković, Željko. 2007. On the locality and motivation of Move and Agree: An even more minimal theory. Linguistic Inquiry 38:589–644.

Branan, Kenyon. 2018. Attraction at a distance: Ā-movement and case. Linguistic Inquiry 49:409–440.

Bruening, Benjamin. 2001. Syntax at the edge: Cross-clausal phenomena and the syntax of Passamaquoddy. Doctoral dissertation, Massachusetts Institute of Technology.

Bruening, Benjamin. 2005. The Algonquian inverse is syntactic: Binding in Passamaquoddy. Ms., University of Delaware .

Bruening, Benjamin. 2013. By phrases in passives and nominals. Syntax 16:1–41.

Bruening, Benjamin. 2014. Defects of defective intervention. Linguistic Inquiry 45:707–719.

Burzio, Luigi. 1986. Italian syntax: A Government-Binding approach. Dordrecht: Springer.

Butt, Miriam, & Jafar Rizvi. 2010. Tense and aspect in urdu. In Layers of aspect, ed. P. Cabredo-Hofherr & B. Laca, 43–66. Stanford: CSLI Publications.

Cardinaletti, Anna. 1997. Agreement and control in expletive constructions. Linguistic Inquiry 28:521–533.

Cardinaletti, Anna. 2001. A second thought on *emarginazione*: destressing vs. 'right dislocation'. In Current studies in Italian syntax. Essays offered to Lorenzo Renzi, ed. Guglielmo Cinque & Gian Paolo Salvi, 117–135. Amsterdam: Elsevier Publishers.

Chomsky, Noam. 1973. Conditions on transformations. In A festschrift for Morris Halle, ed. Stephen R. Anderson & Paul Kiparsky, 232–286. New York: Holt, Reinehart and Winston.

Chomsky, Noam. 1995a. Bare phrase structure. In Government and binding theory and the minimalist program, ed. Gert Webelhuth, 383–440. Oxford: Blackwell.

Chomsky, Noam. 1995b. The Minimalist Program. Cambridge, Mass.: MIT Press.

Chomsky, Noam. 2000. Minimalist inquiries: The framework. In Step by step, ed. Roger Martin, David Michaels, & Juan Uriagereka, 89–155. Cambridge, Mass.: MIT Press.

Chomsky, Noam. 2001. Derivation by phase. In Ken Hale. A life in language, ed. Michael Kenstowicz, 1–52. Cambridge, Mass.: MIT Press.

Cinque, Guglielmo. 1988. On *si* constructions and the theory of *arb*. Linguistic Inquiry 19:521–581.

Cocchi, Gloria. 1995. La selezione dell'ausiliare. Padova: Unipress.

Collins, Chris. 2005. A smuggling approach to the passive in English. Syntax 8:81–120.

Coon, Jessica, & Stefan Keine. 2020. Feature gluttony. Linguistic Inquiry 52:655–710.

Costa, João. 2000. Word order and discourse-configurationality in European Portuguese. In Portuguese syntax: New comparative studies, ed. João Costa, 94–115. Oxford: Oxford University Press.

D'Alessandro, Roberta. 2004. Impersonal *si* constructions: Agreement and interpretation. Doctoral dissertation, Universität Stuttgart.

D'Alessandro, Roberta. 2017a. Agreement in Italian impersonal *si* constructions: A derivational analysis. Revista da ABRALIN 1.

D'Alessandro, Roberta. 2017b. When you have too many features: auxiliaries, agreement, and clitics in Italian varieties. Glossa 2(1).

D'Alessandro, Roberta, & Ian G. Roberts. 2010. Past participle agreement in Abruzzese: split auxiliary selection and the null-subject parameter. Natural Language & Linguistic Theory 28:41–72.

De Alencar, Leonel Figuiredo, & Carmen Kelling. 2005. Are reflexive constructions transitive or intransitive? Evidence from German and Romance. In Proceedings of the LFG05 Conference, ed. Miriam Butt & Tracy Holloway King, 1–20.

Embick, David. 2004. Unaccusative syntax and verbal alternations. In The unaccusativity puzzle. Explorations of the Syntax-Lexicon interface, ed. Artemis Alexiadou, Elena Anagnostopoulou, & Martin Everaert, 137–158. Oxford: Oxford University Press.

Ershova, Ksenia. 2024. Phasehood as defective intervention: Possessor extraction and selective DP islandhood in West Circassian. Syntax 1–46.

Fanselow, Gisbert. 1991. Minimale Syntax. Unpublished habilitation thesis, Universität Passau.

Fischer, Silke. 2004. Towards an optimal theory of reflexivization. Doctoral dissertation, Universität Tübingen.

Fritzsche, Rosa. to appear. Anti-local agreement. Doctoral dissertation, Universität Leipzig.

Gallego, Ángel J. 2013. Object shift in Romance. Natural Language & Linguistic Theory 31:409–451.

Georgi, Doreen. 2014. Opaque interactions of Merge and Agree: On the nature and order of elementary operations. Doctoral dissertation, University of Leipzig.

Grimshaw, Jane Barbara. 1982. On the lexical representation of Romance reflexive clitics. In The mental representation of grammatical relations, ed. Joan Bresnan, 87–146. Cambridge, Mass.: MIT Press.

Hale, Ken, & Samuel Jay Keyser. 2002. Prolegomenon to a theory of argument structure. Cambridge, Mass.: MIT press.

Hale, Kenneth, & Samuel J. Keyser. 1993. On argument structure and the lexical expression of syntactic relations. In The View from Building 20: Essays in Linguistics in Honor of Sylvain Bromberger, ed. Kenneth Hale & Samuel J. Keyser, 53–110. Cambridge, Mass.: MIT Press.

Halle, Morris, & Alec Marantz. 1993. Distributed Morphology and the pieces of inflection. In The View from Building 20: Essays in Linguistics in Honor of Sylvain Bromberger, ed. Kenneth Hale & Samuel J. Keyser, 111–176. Cambridge, Mass.: MIT Press.

Halpert, Claire. 2012. Argument licensing and agreement in Zulu. Doctoral dissertation, Massachusetts Institute of Technology.

Harizanov, Boris, & Vera Gribanova. 2019. Whither head movement? Natural Language & Linguistic Theory 37:461–522.

Harley, Heidi, & Rolf Noyer. 1999. Distributed Morphology. Glot international 4:3–9.

Heck, Fabian. 2016. Non-monotonic derivations. Habiliation thesis, Universität Leipzig.

Heck, Fabian, & Gereon Müller. 2006. Extremely local optimization. In Proceedings of WECOL 34, ed. Erin Bainbridge & Brian Agbayani, 170–182.

Heinat, Fredrik. 2006. Probes, pronouns and binding in the minimalist program. Doctoral dissertation, Lund University.

Hicks, Glyn. 2009. The derivation of anaphoric relations. Amsterdam: John Benjamins Publishing Company.

Hiraiwa, Ken. 2001. Multiple Agree and the defective intervention constraint in Japanese. In Proceedings of the First HUMIT Student Conference in Language Research (HUMIT 2000), ed. Ora Matushansky, Albert Costa, Javier Martin-Gonzalez, Lance Nathan, & Adam Szczegielniak, MIT Working Papers in Linguistics, 67–80. Cambridge, Mass.

Holmberg, Anders, & Thorbjörg Hróarsdóttir. 2003. Agreement and movement in Icelandic raising constructions. Lingua 113:997–1019.

Hornstein, Norbert. 2009. A theory of syntax: Minimal operations and Universal Grammar. New York: Cambridge University Press.

Kalin, Laura Mennen. 2014. Aspect and argument licensing in neo-aramaic. Doctoral dissertation, University of California, Los Angeles.

Kalin, Laura Mennen, & Coppe van Urk. 2015. Aspect splits without ergativity. Natural Language & Linguistic Theory 33:659–702.

Kayne, Richard S. 1975. French syntax: The transformational cycle. Cambridge, Mass.: MIT press.

Kayne, Richard S. 1989. Facets of past participle agreement in Romance. In Dialect variation and the theory of grammar, ed. Paola Benincà, 85–103. Dordrecht: Foris.

Kayne, Richard S. 1993. Toward a modular theory of auxiliary selection. Studia linguistica 47:3–31.

Koizumi, Masatoshi. 1994. Layered specifiers. In Proceedings of NELS 24, ed. Mercè Gonzàlez, 255–269. Amherst: GLSA.

van Koppen, Marjo. 2005. One probe–two goals: Aspects of agreement in Dutch dialects. Doctoral dissertation, Leiden University.

Kučerová, Ivona. 2016. Long-distance agreement in Icelandic: locality restored. The Journal of Comparative Germanic Linguistics 19:49–74.

Lahne, Antje. 2008. Excluding SVO in ergative languages: A new view on Mahajan's generalisation. In Varieties of competition, ed. Fabian Heck, Gereon Müller, & Jochen Trommer, Linguistische Arbeitsberichte, 65–80. Universität Leipzig.

Lahne, Antje. 2012. Specificity-driven syntactic derivation. In Ways of structure building, ed. Myriam Uribe-Etxebarria & Vidal Valmala, 271–296. Oxford: Oxford University Press.

Laka, Itziar. 1993. Unergatives that assign ergative, unaccusatives that assign accusative. In MIT Working Papers in Linguistics 18: Papers on case and agreement, ed. Jonathan David Bobaljik & Collin Phillips, 149–172. Cambridge, Mass.: MITWPL.

Ledgeway, Adam. 2019. Parameters in the development of Romance perfective auxiliary selection. In Historical linguistics 2015 : Selected papers from the 22nd International Conference on Historical Linguistics, ed. Michela Cennamo & Claudia Fabrizio, 343–384. Amsterdam: John Benjamins Publishing Company.

Legate, Julie Anne. 2003. Some interface properties of the phase. Linguistic Inquiry 34:506–515.

Legate, Julie Anne. 2005. Phases and cyclic agreement. In Perspectives on phases, ed. Martha McGinnis & Norvin Richards, MIT Working Papers in Linguistics, 147–156. Cambridge, Mass.

Legate, Julie Anne. 2008. Morphological and abstract case. Linguistic Inquiry 39:55–101.

Legate, Julie Anne. 2014. Voice and *v*: Lessons from Acehnese. Cambridge, Mass.: MIT Press.

Longenbaugh, Nicholas. 2019. On expletives and the agreement-movement correlation. Doctoral dissertation, Massachusetts Institute of Technology.

Loporcaro, Michele. 1998. Sintassi comparata dell'accordo participiale romanzo. Torino: Rosenberg & Sellier.

Loporcaro, Michele. 2007. On triple auxiliation in Romance. Linguistics 45:173–222.

Mahajan, Anoop. 1989. Agreement and agreement projections. In Functional heads and clause structure. mit working papers in linguistics, ed. Itziar Laka & Anoop Mahajan, 10, 217–252. Cambridge, Mass.

Mahajan, Anoop Kumar. 1990. The a/a-bar distinction and movement theory. Doctoral dissertation, Massachusetts Institute of Technology.

Manetta, Emily. 2011. Peripheries in Kashmiri and Hindi-Urdu: The syntax of discourse-driven movement. Amsterdam: John Benjamins Publishing Company.

Manzini, Maria Rita. 1986. On italian *si*. In Syntax and semantics: The syntax of pronominal clitics, ed. Hagit Borer, 241–262. New York: Academic Press.

Manzini, Maria Rita, & Leonardo Maria Savoia. 2005. I dialetti italiani: sintassi delle varietà italiane e romance. Alessandria: Edizioni dell'Orso.

Marantz, Alec. 1984. On the nature of grammatical relations. Cambridge, Mass.: MIT Press.

Marantz, Alec. 1991. Case and licensing. In Proceedings of the Eighth Eastern States Conference on Linguistics, ed. Germán F. Westphal, Benjamin Ao, & HeRahk Chae, 234–253.

Martinović, Martina. 2021. Feature geometry and head splitting in the Wolof clausal periphery. Linguistic Inquiry 1–38.

McCawley, James. 1984. Exploitation of the cyclic principle as a research strategy in syntax. In Sentential complementation, ed. Wim de Geest & Yvan Putseys, 165–183. Dordrecht: Foris.

McCawley, James. 1988. The syntactic phenomena of English. Chicago: University of Chicago Press.

McFadden, Thomas, & Sandhya Sundaresan. 2011. Nominative case is independent of finiteness and agreement. Ms., University of Tromsø .

Mohanan, Tara. 1994. Argument structure in hindi. Stanford: CSLI Publications.

Moravcsik, Edith A. 1974. Object-verb agreement. Working papers on language universals 15:25–140.

Mulders, Iris. 1997. Mirrored specifiers. Linguistics in the Netherlands 14:135–146.

Müller, Gereon. 2004. Phrase impenetrability and wh-intervention. In Minimality effects in syntax, ed. Arthur Stepanov, Gisbert Fanselow, & Ralf Vogel, 289–325. Berlin: Mouton de Gruyter.

Müller, Gereon. 2009. Ergativity, accusativity, and the order of Merge and Agree. In Explorations of phase theory: Features and arguments, ed. Kleanthes K. Grohmann, 269–308. Berlin: Mouton de Gruyter.

Müller, Gereon. 2010. On deriving CED effects from the PIC. Linguistic Inquiry 41:35–82.

Müller, Gereon. 2011. Constraints on displacement: A phase-based approach. Amsterdam: John Benjamins Publishing Company.

Murphy, Andrew. 2018. Dative intervention is a gang effect. The Linguistic Review 35:519–574.

Murugesan, Gurujegan. 2022. Deriving the anaphor–agreement effect and the violations of it. Syntax 25:39–83.

Myler, Neil. 2018. Complex copula systems as suppletive allomorphy. Glossa: a journal of general linguistics 3.

Myler, Neil Joseph. 2014. Building and interpreting possession sentences. Doctoral dissertation, New York University.

Müller, Gereon. 2021. Constraints on grammatical dependencies. In A companion to Chomsky, ed. Nicholas Allott, Terye Lohndal, & Georges Rey. Hoboken: Wiley Blackwell.

Nevins, Andrew. 2011. Multiple Agree with clitics: Person complementarity vs. omnivorous number. Natural Language & Linguistic Theory 29:939–971.

Nunes, Jairo. 2019. Edge features and phase head allomorphy. Talk given at GLOW42 Talk given at GLOW42, 2019.

Ordóñez, Francisco. 1998. Word order and clause structure in Spanish and other Romance languages. Doctoral dissertation, City University of New York, New York.

Perlmutter, David M. 1978. Impersonal passives and the unaccusative hypothesis. In Proceedings of the fourth annual meeting of the Berkeley Linguistics Society, ed. Farrell Ackerman Christine Chiarello Orin D. Gensler John Kingston Eve E. Sweetser Henry Thompson Jeri J. Jaeger, Anthony C. Woodbury & Kenneth W. Whistler, 157–191. Berkeley.

Pesetsky, David. 1989. Language-particular processes and the earliness principle. Ms., Massachusetts Institute of Technology .

Pesetsky, David. 2021. Exfoliation: towards a derivational theory of clause size. Ms., Massachusetts Institute of Technology. Available at lingbuzz/004440 .

Pesetsky, David, & Esther Torrego. 2001. T-to-C movement: Causes and consequences. In Ken Hale. A Life in Language, ed. M. Kenstowicz, 355–426. Cambridge, Mass.: MIT Press.

Pesetsky, David, & Esther Torrego. 2007. The syntax of valuation and the interpretability of features. In Phrasal and clausal architecture: syntactic derivation and interpretation. in honor of Joseph E. Emonds, ed. Simin Karimi, Vida Samiian, & Wendy Wilkins, 262–294. Amsterdam: John Benjamins Publishing Company.

Pineda, Anna. 2014. What lies behind dative/accusative alternations in Romance. In RLLT 2012, ed. S. Marzo & K. Lahousse, 123–139. Amsterdam: John Benjamins Publishing Company.

Preminger, Omer. 2011. Asymmetries between person and number in syntax: a commentary on Baker's SCOPA. Natural Language & Linguistic Theory 29:917–937.

Preminger, Omer. 2014. Agreement and its failures. Cambridge, Mass.: MIT Press.

Puškar, Zorica. 2018. Interactions of gender and number agreement: Evidence from Bosnian/Croatian/Serbian. Syntax 21:275–318.

Pylkkänen, Liina. 2008. Introducing arguments. Doctoral dissertation, Massachusetts Institute of Technology.

Rackowski, Andrea, & Norvin W. Richards. 2005. Phase edge and extraction: A Tagalog case study. Linguistic Inquiry 36:565–599.

Radkevich, Nina. 2017. Two types of split ergativity= two sides of the same coin. Talk given at LAGB Annual Meeting, University of Kent .

Radkevich, Nina, & Lauren Eby Clemens. 2013. Person hierarchy and structure dependence in Lak agreement. Talk given at WCCFL 31 .

Raposo, Eduardo. 1987. Case theory and Infl-to-Comp: The inflected infinitive in European Portuguese. Linguistic Inquiry 18:85–109.

Reuland, Eric. 2001. Primitives of binding. Linguistic Inquiry 32:439–492.

Rezac, Milan. 2004. Elements of cyclic syntax: Agree and Merge. Doctoral dissertation, University of Toronto.

Richards, Marc. 2005. Object shift and scrambling in North and West Germanic: A case study in symmetrical syntax. Doctoral dissertation, University of Cambridge.

Richards, Marc. 2008. Quirky expletives. In Agreement restrictions, ed. Roberta d'Alessandro, Susann Fischer, & Gunnar H. Hrafnbjargarson, 181–213. Berlin: de Gruyter.

Richards, Norvin W. 1997. What moves where when in which languages? Doctoral dissertation, Massachusetts Institute of Technology.

Richards, Norvin W. 1998. The Principle of Minimal Compliance. Linguistic Inquiry 29:599–629.

Richards, Norvin W. 2013. Lardil "case stacking" and the timing of case assignment. Syntax 16:42–76.

Rizzi, Luigi. 1982. Issues in Italian Syntax. Dordrecht: Foris.

Rizzi, Luigi. 1990. Relativized minimality. Cambridge, Mass.: MIT Press.

Roberts, Ian G. 2010. Agreement and head movement: Clitics, incorporation, and defective goals. Cambridge, Mass.: MIT Press.

Rooryck, Johan, & Guido J. Vanden Wyngaerd. 2011. Dissolving binding theory. New York: Oxford University Press.

Rudin, Catherine. 1988. On multiple questions and multiple wh fronting. Natural Language & Linguistic Theory 6:445–501.

Sabel, Joachim. 1998. Principles and parameters of wh-movement. Habilitation thesis, Universität Frankfurt.

Schwarze, Christoph. 1998. A lexical-functional analysis of Romance auxiliaries. Theoretical Linguistics 24:83–105.

Sigurðsson, Halldór Ármann. 1989. Verbal syntax and case in Icelandic in a comparative GB approach. Doctoral dissertation, University of Lund.

Sigurðsson, Halldór Ármann. 1991. Icelandic case-marked PRO and the licensing of lexical arguments. Natural Language & Linguistic Theory 9:327–363.

Sigurðsson, Halldór Ármann. 1996. Icelandic finite verb agreement. Working Papers in Scandinavian Syntax 57:1–46.

Sigurðsson, Halldór Ármann. 2006. The nominative puzzle and the low nominative hypothesis. Linguistic Inquiry 37:289–308.

Sigurðsson, Halldór Ármann, & Anders Holmberg. 2008. Icelandic dative intervention: Person and number are separate probes. In Agreement restrictions, ed. Roberta d'Alessandro, Susann Fischer, & Gunnar H. Hrafnbjargarson, 181–213. Berlin: de Gruyter.

Sorace, Antonella. 2000. Gradients in auxiliary selection with intransitive verbs. Language 76:859–890.

Sportiche, Dominique. 2014. Assessing unaccusativity and reflexivity: Using focus alternatives to decide what gets which θ-role. Linguistic Inquiry 45:305–321.

Svenonius, Peter. 2006. Case alternations and the Icelandic passive and middle. In Passives and impersonals in European languages, ed. Katrin Hieetam Elsi Kaiser Satu Manninen, Diane Nelson & Virve Vihman. Amsterdam: John Benjamins Publishing Company.

Taraldsen, Knut Tarald. 1995. On agreement and nominative objects in Icelandic. In Studies in comparative germanic syntax, ed. H. Haider, S. Olsen, & S. Vikner, 307–327. Dordrecht: Springer.

Ura, Hiroyuki. 2000. Cheking theory and grammatial funtions in Universal Grammar. New York: Oxford University Press.

van Urk, Coppe. 2015. A uniform syntax for phrasal movement: A case study of dinka bor. Doctoral dissertation, Massachusetts Institute of Technology.

van Urk, Coppe. 2020. Successive cyclicity and the syntax of long-distance dependencies. Annual Review of Linguistics 6:111–130.

van Urk, Coppe, & Norvin W. Richards. 2015. Two components of long-distance extraction: Successive cyclicity in Dinka. Linguistic Inquiry 46:113–155.

Ussery, Cherlon. 2009. Optionality and variability: Syntactic licensing meets morphological Spell-out. Doctoral dissertation, University of Massachusetts, Amherst, MA.

Ussery, Cherlon. 2017. Dimensions of variation. Agreement with nominative objects in Icelandic. In Syntactic variation in Insular Scandinavian, ed. Hjalmar P. Petersen Höskuldur Thráinsson, Caroline Heycock & Zakaris Svabo Hansen, 166–197. Amsterdam: John Benjamins Publishing Company.

Vainikka, Anne, & Pauli Brattico. 2014. The Finnish accusative: Long-distance case assignment under agreement. Linguistics 52:73–124.

Vicente, Luis. 2007. The syntax of heads and phrases: A study of verb (phrase) fronting. Utrecht: LOT.

Wurmbrand, Susi. 2012. Parasitic participles in Germanic: evidence for the theory of verb clusters. Taal en Tongval 129–156.

Zaenen, Annie, Joan Maling, & Höskuldur Thráinsson. 1985. Case and grammatical functions: The Icelandic passive. Natural Language & Linguistic Theory 3:441–483.

Zubizarreta, Maria Luisa. 1998. Prosody, focus, and word order. Cambridge, Mass.: MIT Press.

6 Appendix: Nested Agree and other principles

Many other principles similar to Nested Agree have already been proposed in previous literature: *Maximize Matching Effects (MME)* (Chomsky 2001: 15), *General Specificity Principle (GSP)* (Lahne 2012: 2), *Multitasking (MT)* (van Urk & Richards 2015: 132), *Feature Maximality (FM)* (Longenbaugh 2019: 21), and *Economy condition on multiple probe satisfaction (ECoMPS)* (Pesetsky 2021: 28).

Maximize Matching Effects (Chomsky 2001: 15) states that when two heads enter into a relation, they should check together as many features as possible. It prescribes that each probe must target the highest matching goal, as soon as possible. Under this perspective, it can be considered as a consequence of the Earliness principle (Pesetsky 1989), which prescribes that an operation between a probe and a goal should happen as soon as possible. Note that MME is not a specificity-driven principle because it does not favor the operation that satisfies more features at the same time. In fact, it strictly obeys minimality: if there is a higher goal for a probe, this should be chosen instead of a lower one, even when the lower goal would satisfy several probes at once.

Given the multiple probing effect of Nested Agree, the reader may wonder whether this novel principle can be reduced to a maximizing principle such as MME. This parallelism seems particularly appealing in those scenarios where several probes are satisfied by a single goal, which are most of the cases discussed in this article. However, it is exactly for its relationship with minimality that MME cannot allow for the configurations that Nested Agree tolerates. In several points of the article, I have briefly noted how MME cannot account for the discussed data (see section 3.3, 4.1.1, and 4.1.2). Nested Agree is also partly opposite to MME: depending on the feature ordering, it might allow a probe-feature to skip the most local goal feature. Hence, MME or Earliness do not constitute valid alternatives to Nested Agree. The sequentiality that characterizes Nested Agree (which is an effect of feature ordering) makes it unique compared to previous principles of this type. Feature ordering is the core component of Nested Agree, which makes it a new proposal profoundly different from MME. Other effects of Nested Agree, such as the truncation of the search space, follow as consequences of probing being subject to other conditions, such as no-backtracking, and no upward cyclic expansion.

As far as the other principles mentioned above are concerned (GSP, MT, FM, and ECoMPS), Nested Agree differs from them for several reasons, which all stem from feature ordering. Firstly, it is not specificity-driven. Secondly, it might allow a probe-feature to skip the most local goal-feature in favor of a non-local goal-feature. Thirdly, it is not transderivational. Fourthly, it has the advantage of capturing cross-linguistic variation via feature re-ordering. I will now discuss these points in detail.

The main difference between Nested Agree and the principles mentioned above is that the latter are *specificity-driven*, whereas the former is not. In specificity-driven derivations, elements that are more "specific" in their featural inventory (meaning that they carry more features, or more articulated values of these features) are preferred by the probe(s) over less specific elements. When two derivations are possible, these principles force the one that involves fewer operations to take place. On the contrary, Nested Agree does not favor the most specific goal in its c-command domain. It does not require that two heads match as many features as possible, but rather that an existing dependency is exploited as much as possible as the first step of any new search.

As independent of specificity, Nested Agree can also account for anti-specificity and anti-locality effects, which cannot be covered by these principles. Due to feature ordering, it might allow a probe-feature to skip the most local goal-feature in favor of a non-local goal-feature. This happens, for instance, when the higher goal is more specific than the lower one, but lacks the feature that is probed first (e.g., the higher goal bears the features α and β, the lower goal bears γ, and the probe for γ precedes the probes for α and β). In other words, Nested Agree can lead to a kind of "relativized anti-locality" induced by feature ordering. More research is needed to investigate this aspect, but I believe that many cases of anti-locality could be modeled as an effect of feature ordering (see sections 4.2.1, 4.3.1 for potential case studies). Crucially, data of this kind cannot be modeled by any of the principles mentioned above.

The status of these principles in relation to minimality is also different. Nested Agree is subject to minimality, but it can be used to model anti-minimality violations, as shown in this paper. The principles mentioned above cannot do so, because either no anti-minimality effects are tolerated at all (ECoMPS), or minimality is not addressed as a relevant issue (MT).

Another difference between Nested Agree and at least some of these principles (MT, ECoMPS, and GSP) is that the latter are transderivational constraints: a derivation is evaluated in contrast with other possible derivations and is preferred if it leads to the satisfaction of more operations. However, this possibility should be ruled out in a derivational approach, such as the one adopted in the present article and advocated by Minimalism in general. For each step of the derivation, only one option should be possible. Nested Agree has a different computational status than these principles: it is exactly derivational. It does not compare competing derivations. It only determines that, as an obligatory consequence of feature ordering, a probe must start its search from the last inspected head (this information is available because the dependency between two heads is stored, as prescribed in the Agree-Link theory, see section 2).

An important question is whether Nested Agree has the same empirical coverage of these principles. I believe the answer is positive. In particular, specificity effects can be rephrased as the result of a feature hierarchy. Nested Agree offers a new way of interpreting these phenomena from a derivational perspective. For reasons of space, I will only discuss the two case studies that led to the formulation of MT and ECoMPS.

Multitasking (van Urk & Richards 2015: 132) has been proposed to account for cases of competition for movement between two DPs. The crucial data comes from Dinka. In this language, Spec,*v* must be filled by one constituent. In ditransitive clauses, there are two options: either the indirect object or the direct object moves to Spec,*v*. However, if the object is a wh-phrase, only this phrase can move to Spec,*v* (instead of the higher indirect object). van Urk & Richards (2015) argue that movement of wh-obj is performed because it satisfies both probes [$*$wh$*$] and [$*$case$*$] on *v* at once (the former triggers wh-movement, the latter EPP movement to

Spec,v). The derivation that satisfies most probes with fewer operations is enforced by MT. This principle excludes the other competing derivation where each item moves to satisfy a separate probe.

Nested Agree can model these data by assuming that in Dinka the order of the probe-features on v is [$*$wh$*$] $\succ$ [$*$case$*$].[34] The wh-phrase in the object position is the goal of the first probe [$*$wh$*$]; Nested Agree restricts the domain of the second operation [$*$case$*$] to the same goal. Thus, Nested Agree "translates" the priority of Ā-probing into feature ordering, thereby also eliminating any minimality problem (which instead affects the MT-analysis, although this is not discussed by van Urk & Richards 2015: 122). A confirmation of the primacy of the Ā-probe over the case/EPP-probe comes from Dinka causatives (van Urk 2015: 164). The causative verb requires that a DP from inside its complement (i.e., the embedded subject or object) move to its Spec,v position. If one of the embedded arguments is a wh-phrase, this one must move to Spec,v. As in ditransitives, in this configuration wh-movement blocks movement of other items to this position. This means that wh-probing applies first. The primacy of the Ā-probe can also be seen at work in long-distance dependencies: the moving wh-phrase satisfies the V2 property of all intervening clauses (van Urk 2020: 119).

The Economy condition on multiple probe satisfaction (Pesetsky 2021: 28) was proposed for a similar case of competing derivations. In English, the verb *seem* allows for free alternation between raising and expletive insertion. However, when the embedded subject is a wh-phrase, only the raising-derivation is possible. Pesetsky (2021: 26-29) proposes that wh-raising is the only option because a single element satisfies both probes [$*$wh$*$] and [$*$EPP$*$] on v, while in the expletive-derivation two independent movement operations would be needed. This less economical option is blocked by ECoMPS. Nested Agree can account for these data with the feature ordering [$*$wh$*$] $\succ$ [$*$EPP$*$]. The wh-probe targets the wh-phrase, and the EPP feature must do so as well.

Not only does Nested Agree have the same empirical coverage as the principles discussed

[34]Here I simplify the structure of the Merge feature. I represent it as an Agree feature, following van Urk & Richards (2015); Pesetsky (2021), based on the idea that movement is feature-driven. More precisely, each Merge feature contains an Agree-component [$*$F$*$] and a structure-building component [•F•]. In the analysis sketched below, the Agree-components of both Merge features go first in this order: [$*$wh$*$] $\succ$ [$*$EPP$*$]/[$*$case$*$]. They both agree with the wh-phrase. This is then raised via [•wh•]. In general, more case studies are needed to explore the relation between movement and Nested Agree (see sections 4.2.3, 4.3.2 for some exploratory case studies).

in this appendix. It can also derive the case studies presented in this paper, unlike the other principles. For instance, none of these can account for auxiliary selection in Italian because of two main obstacles. Firstly, there is a case where both potential goals are equally specific. In unaccusative clauses (section 3.4.2), *v* bears [Infl], and DP [π]. Since specificity does not play a role here, there is no way in which both probes on Perf should target *v* and skip the DP. Secondly, person-driven auxiliary selection (section 3.5) cannot be derived, since *v* would always be favored by Perf in a specificity-driven scenario.

Thus, Nested Agree cannot be reduced to any of these principles. Nonetheless, it can be rephrased as the combination of the following ingredients: (i) feature ordering, (ii) MME (or one of the principles mentioned above), (iii) the *Principle of Minimal Compliance (PMC)* (Richards 1998: 601).[35] Feature ordering ensures sequentiality and its consequences (no specificity-driven derivation, no transderivationality, possible emergence of apparent anti-minimality configurations); the maximizing or specificity-driven principle enables multiple probing; the PMC tolerates eventual minimality violations if these arise after a first well-formed Agree-operation. Further studies should establish if Nested Agree emerges as an ephiphenomenon from the combination of these various components, rather than being a completely new principle of syntax. Regardless of this question, the fundamental result of this paper is that various independent phenomena cannot be derived by any existing principle taken in isolation, but only by this novel combination of principles, which I have called Nested Agree.

[35]It is not easy to prove that Nested Agree has the same empirical coverage as the PMC because these two principles are very different. The PMC regulates the application of other syntactic constraints (e.g., a Principle C violation can be tolerated if Principle C has been respected by another "relevant" dependency). At first sight, the PMC seems to account for auxiliary selection because it allows for a non-local Agree relation once a well-formed Agree dependency is established. However, without feature ordering the PMC cannot let a probe skip the higher DP and carry out a non-local operation. In addition, without multiple probing the PMC has no way to let a probe target the same goal twice. Hence, auxiliary selection in Standard Italian cannot be derived by the PMC. Moreover, cross-linguistic variation cannot be explained by the PMC either.